

Genetically Modified Organisms, Cloning Techniques, and Developmental & Spatial Gene Expression - a complete university teaching module

For BS/MS Molecular Biology, Biotechnology, Genetics, Bioinformatics,
Biomedical Sciences

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All datasets used in this package are simulated teaching data. Wet-lab and animal procedures described here are conceptual; institutional biosafety approval and SOPs govern all practice.

Table of Contents

Placeholder for table of contents	0
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Module 1 — Introduction to Genetic Engineering

Course: GMOs, Cloning Techniques, and Developmental & Spatial Gene Expression

Level: Beginner (foundation for all later modules)

Navigation: [↑ Course Home](#) · [← \(start of course\)](#) · [→ Next Topic](#)

■ [Related Lab: Lab 01 — DNA Cloning Design](#) · ■ [Data Exercise: Cloning data](#) · ■ [Assessment: MCQs](#) · ■ [Cheat Sheet](#) · ■ [FAQ](#)

Learning objectives

After this module you should be able to:

1. Define **genetic engineering**, **recombinant DNA (rDNA) technology**, and **molecular cloning**, and place them inside the broader field of **biotechnology**.
2. Distinguish the terms **GMO**, **transgenic**, **cisgenic**, **gene-edited**, and **genetically engineered**, and explain why different jurisdictions draw the boundaries differently.
3. Recount the historical milestones that produced modern genetic engineering, from classical genetics to CRISPR and spatial/single-cell genomics.
4. Explain *why* organisms are genetically modified: research, medicine, agriculture, and industry.
5. Distinguish **molecular cloning** (making a recombinant DNA molecule) from **expression engineering** (making that molecule produce a gene product in a host).

1. What is genetic engineering?

Genetic engineering is the direct, deliberate modification of an organism's genetic material using laboratory techniques, in ways that would not necessarily occur through conventional breeding or natural recombination.

Three ideas sit at its core:

- **Precision of intent** — a defined change (a new gene, a repaired mutation, a disrupted reading frame) is designed before the experiment, not merely selected afterwards.
- **Molecular tools** — enzymes (restriction endonucleases, ligases, polymerases, recombinases, Cas nucleases) and DNA vehicles (vectors) do the cutting, joining and delivery.
- **Heritable or controlled change** — the engineered DNA must either be maintained by the cell (on a plasmid, integrated into a chromosome) or be present transiently long enough to act.

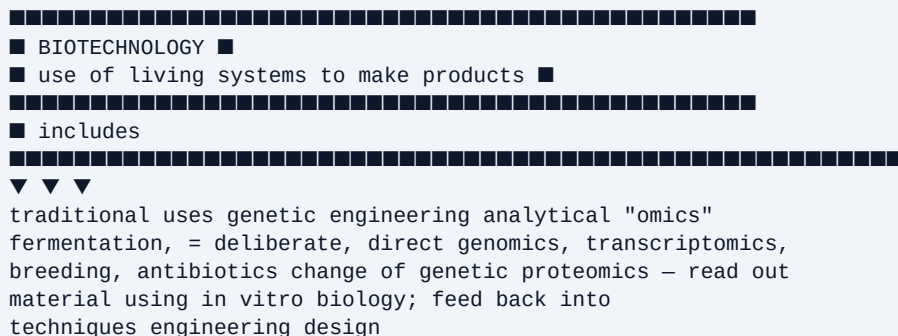


Figure 1.1 — Where genetic engineering sits within biotechnology.

1.1 Recombinant DNA technology

Recombinant DNA (rDNA) technology is the set of laboratory methods for joining DNA from more than one source into a DNA molecule that can be propagated in a host cell. The product is a **recombinant DNA molecule** — most commonly a plasmid carrying an insert (a "recombinant plasmid").

The canonical experiment — Cohen, Chang, Boyer and Helling (1973) — cut two plasmids with a restriction enzyme, ligated them together, and introduced the hybrid plasmid into *Escherichia coli*, which then expressed a gene from the other plasmid. This single demonstration established the recombinant-DNA paradigm: **cut** → **join** → **propagate** → **select**.

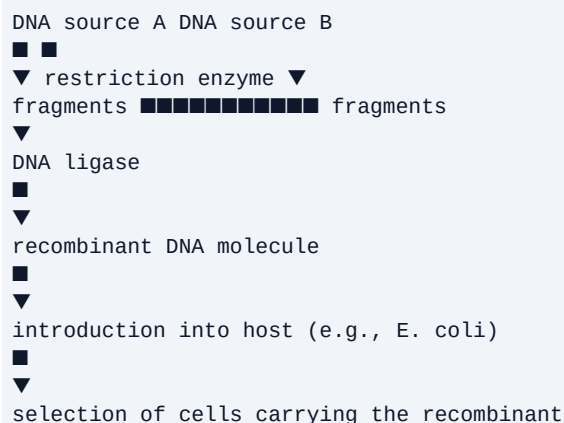


Figure 1.2 — The recombinant DNA cycle.

1.2 Molecular cloning vs expression engineering

- **Molecular cloning (DNA cloning):** constructing and amplifying a defined DNA molecule — the *blueprint* stage. Output: a verified plasmid or DNA fragment. (Detailed in Modules 2–5.)
- **Expression engineering:** designing constructs so that the DNA is **transcribed and translated** in the intended host, at the intended level, time and place — the *manufacturing* stage. Output: a protein or RNA product, or a visible reporter signal. (Modules 5, 8–11.)

A construct that clones beautifully but expresses nothing is a common beginner outcome; the difference lies in regulatory elements (promoters, terminators, translation-initiation contexts), which Module 5 treats in depth.

2. The vocabulary of modification — precise distinctions

These terms are often used loosely. Use them precisely; examiners and regulators certainly do.

Term	Definition	Key discriminator
Genetically modified organism (GMO)	Any organism whose genetic material has been altered by human intervention using genetic-engineering techniques	Umbrella term; definition is <i>regulatory</i> , and varies by jurisdiction
Genetically engineered (GE)	Synonymous with GMO in most scientific usage	Emphasis on the technique rather than the result
Transgenic	Carries foreign DNA from a different species (e.g., a mouse expressing a human gene; Bt corn carrying a bacterial <i>cry</i> gene)	DNA crosses species boundaries
Cisgenic	Carries DNA from the same species (or a very closely related, cross-compatible species) — e.g., a potato variety with a resistance gene from another potato variety	DNA stays within the species' gene pool
Gene-edited	Carries a change made by a targeted editor (e.g., CRISPR/Cas); the change may be a small deletion, a base change, or an inserted sequence — which may or may not be transgenic	The <i>process</i> is targeted editing; the product may be indistinguishable from a natural variant
Knockout	A gene deliberately inactivated	Usually gene-edited or homologous-recombination based

Term	Definition	Key discriminator
Knock-in	A defined sequence inserted at a specific locus	Typically CRISPR/HDR or recombinase-mediated
Stable modification	The engineered DNA is inherited through cell divisions (or generations) — plasmid maintained under selection, or integrated into the genome	Persistence across generations
Transient modification	DNA/RNA is introduced but not replicated or integrated ; expression fades over hours–days (e.g., plasmid transfection of cultured cells, mRNA injection into embryos)	No genomic integration; no inheritance

Reading the table critically: whether a herbicide-tolerant crop produced by CRISPR base editing counts as a "GMO" is a *legal* question with different answers in different countries, not a settled scientific one. The scientific descriptors (what DNA is present, what change was made) are unambiguous; the regulatory labels are not.

3. Why genetically modify organisms?

Purpose	Typical aims	Examples
Basic research	Assign function to genes; build disease models; visualize cells and molecules	GFP-tagged proteins; knockout mice; developmental reporters in zebrafish
Medicine	Produce therapeutic proteins; gene therapy; cell therapy	Recombinant human insulin (1982, first approved rDNA drug); CAR-T cells; mRNA vaccines
Agriculture	Pest resistance, disease resistance, herbicide tolerance, improved nutrition, reduced spoilage	Bt cotton/corn; papaya ringspot-virus-resistant papaya; Golden Rice; non-browning apple
Industry	Enzymes, biofuels, bio-based materials	Amylases for starch processing; engineered yeast producing artemisinic acid

The **research** column deserves emphasis for this course: the vast majority of genetic engineering done worldwide is done to *ask questions* — which cell expresses this gene, what does this protein do, what happens when this circuit is broken — rather than to make products.

4. Historical development

Classical Genetics → DNA discovery → Restriction enzymes → Recombinant DNA
 → Molecular cloning → Transgenic organisms → Genome sequencing → CRISPR/Cas
 → Synthetic biology → Spatial and single-cell genomics

Era	Milestone	Why it mattered
1865–1900s	Mendel's laws rediscovered (1900); chromosome theory (early 1900s)	Inheritance becomes a physical science
1928	Griffith's "transforming principle" in <i>Streptococcus pneumoniae</i>	Something in dead bacteria carries heritable information
1944	Avery, MacLeod, McCarty identify the transforming principle as DNA	DNA is the genetic material
1953	Watson & Crick, with Franklin's and Wilkins' X-ray data — DNA double helix	Structure implies copying mechanism
1961–66	Genetic code cracked (Nirenberg, Matthaei, Khorana)	DNA sequence becomes interpretable

Era	Milestone	Why it mattered
1968–70	Discovery and characterization of restriction endonucleases (Meselson & Yuan; Hamilton Smith ; purification of EcoRI); Werner Arber's earlier work on restriction–modification	Molecular "scissors" exist — DNA becomes cuttable at defined sites
1972	Berg & Jackson — first recombinant DNA molecules in vitro	DNA from different sources joined in a test tube
1973	Cohen, Chang, Boyer & Helling — functional recombinant plasmids cloned in <i>E. coli</i>	The cloning paradigm works end to end
1975	Asilomar conference — scientists pause and set rDNA safety guidelines	Self-regulation; origin of modern biosafety frameworks
1977	Sanger & Coulson / Maxam & Gilbert — DNA sequencing; first recombinant human protein expressed in bacteria (somatostatin, then insulin, Genentech)	Sequence reads and expression of human genes in microbes
1978–82	First recombinant human insulin approved (Humulin, 1982)	rDNA technology enters the clinic
1980s	Transgenic mice via pronuclear microinjection (Gordon & Ruddle, 1981–82); "supermouse" growth-hormone mouse (Palmiter & Brinster, 1982); Agrobacterium-mediated plant transformation (Herrera-Estrella et al., 1983); PCR invented (Kary Mullis, 1983; published 1985)	Whole organisms engineered; DNA amplified in vitro
1990s	HGP begins (1990); GFP demonstrated as a genetically encoded fluorescent reporter (Chalfie et al., 1994; Tsien's fluorescent-protein engineering)	See gene expression in living tissue; read whole genomes
1996	First cloned mammal from an adult somatic cell — Dolly the sheep (Wilmut et al., published 1997)	Somatic-cell nuclear transfer; cloning ≠ transgenesis
2000s	HGP draft (2001); RNA interference harnessed (Fire & Mello, 1998; Nobel 2006); first RNAi therapeutics approved (patisiran, 2018)	Gene silencing as a tool and therapy
2010s	CRISPR–Cas9 adapted for genome editing (Jinek et al., 2012; Cong & Mali, 2013; Zhang lab mammalian work)	Programmable, multiplexable genome editing
2016–	Cas9 structure (Nishimasu et al.); base editors (Komor et al., 2016); prime editors (Anzalone et al., 2019)	Editing without double-strand breaks
2020	Nobel Prize in Chemistry to Doudna & Charpentier for CRISPR genome editing	CRISPR becomes a mainstream molecular tool
2013–	Cas9 knockouts in zebrafish, plants, primates; approved gene therapies (e.g., CASGEVY for sickle-cell disease, 2023)	Editing leaves the laboratory
2020–	Synthetic biology circuits, minimal genomes, engineered biosensors	Engineering life as an engineering discipline
2009–2026	Single-cell RNA-seq (Tang et al. 2009; Drop-seq/InDrop 2015), spatial transcriptomics (Ståhl et al., 2016, Nature Methods "Method of the Year 2020"; 10x Genomics Visium; slide-seq; MERFISH; seqFISH+)	Gene expression read in <i>space and time</i> , not just bulk

(Nobel prizes: Arber, Nathans & Smith 1978 — restriction enzymes; Gilbert & Sanger 1980 — sequencing; Mullis 1993 — PCR; Chalfie, Shimomura & Tsien 2008 — GFP; Gurdon & Yamanaka 2012 — reprogramming; Doudna & Charpentier 2020 — CRISPR.)

Two conceptual revolutions to keep separate:

1. **Recombinant DNA revolution (1970s–1990s):** read, copy, and *add* DNA. You can put a gene into an organism, but you usually cannot dictate *where* it lands.
2. **Genome-editing revolution (2010s–):** *edit* DNA at a chosen address. You can delete, replace, or correct a specific sequence in its native chromosomal context.

Modern genetic engineering increasingly combines both: an edited allele plus a transgenic reporter, for instance.

5. The central dogma as an engineering map

DNA ■■■transcription■■■ RNA ■■■translation■■■ PROTEIN

▲ ■ ■
■ ■ ■

where engineers where engineers where engineers
write/edit tune stability, tune folding,
the message splicing, export, localization,
(constructs, localization activity, half-life
genome editing) (UTRs, miRNA sites)

Engineering levers:

- What DNA sequence is present → gene content
- Which regulatory elements flank it → when/where/how much it is expressed
- In what chromosomal context it sits → epigenetic state, variegation
- Whether it is on a plasmid or integrated → transient vs stable

Figure 1.3 — The central dogma read as a list of engineering control points.

6. Self-check questions

1. A rice line carries a maize gene that improves photosynthetic efficiency, inserted by *Agrobacterium* transformation. Classify it: GMO? transgenic? cisgenic? gene-edited?
2. Why is Dolly the sheep an important *cloning* milestone but not a *transgenic* one?
3. Give one example each of a research, medical, agricultural, and industrial use of genetic engineering, and state what the engineered change is in each case.
4. What practical difference did the Asilomar conference make to how molecular biology is done?
5. Why can the same CRISPR-edited organism be regulated as a GMO in one country and as a conventional product in another?

(Answers are discussed in the [Assessment package](../ASSESSMENT/MCQs.md); model answers in the [Answer Key](../ASSESSMENT/Answer-Key.md).)

7. Where this leads

Module 2 dissects the molecular substrate — DNA fragments, vectors and plasmid architecture — and walks the full molecular-cloning workflow stage by stage. Before that, skim the Cheat Sheet vocabulary section and try the Lab 01 design exercise if you already have basic molecular-biology background.

Navigation: ↑ [Course Home](#) · ← (start of course) · → [Next Topic](#)

■ *Related Labs: Lab 01 — DNA Cloning Design · Lab 02 — Restriction Digestion* · ■ *Data: Cloning data* · ■ *Assessment* · ■ *Cheat Sheet* · ■ *FAQ*

1. Define every structural element of a plasmid cloning vector and explain what breaks when it is missing.
2. Distinguish insert, backbone, multiple cloning site, expression cassette, and fusion constructs.
3. Walk the eleven-stage molecular-cloning workflow and, for each stage, state its purpose, inputs, outputs, typical failure modes, and validation.
4. Read a plasmid map confidently.

Component	Job	Consequence if absent/defective
Origin of replication (ori)	Host replication machinery initiates here; copy number is largely set by the ori (e.g., pUC/pMB1-derived high-copy ~500–700, pBR322-derived ~15–20, pSC101-derived low-copy ~5, F-derived single-copy BACs)	No propagation; or runaway/insufficient copy number
Selectable marker	Allows <i>only</i> cells carrying the vector to survive (most commonly antibiotic resistance: ampR, kanR, camR, tetR, specR)	Cannot distinguish transformed from untransformed cells
Multiple cloning site (MCS)	Cluster of unique restriction sites for inserting DNA	Insertion impossible or unidirectional errors common
Promoter	Drives transcription of the insert (in expression vectors) or of a reporter	Clone exists but is never transcribed
Ribosome-binding site / translation-initiation context	Positions the ribosome (prokaryotes: Shine–Dalgarno; eukaryotes: Kozak context)	Transcript made, no protein
Terminator	Stops transcription at a defined point	Read-through transcription destabilizes constructs
Reporter gene	Quantifiable/visible output (GFP, lacZ, luciferase)	Expression invisible
Tag/fusion elements (His $\blacksquare$, FLAG, HA, GST, fluorescent proteins)	Detection or purification of the protein product	Protein made but hard to detect/purify

■■■■■■■■■■■■■■■■■■■■ plasmid (ds circular DNA) ■■■■■■■■■■■■■■■■■■■■
 ■ ■
 ■ ■■■■ selectable marker (e.g., ampR) ■■■■ ■

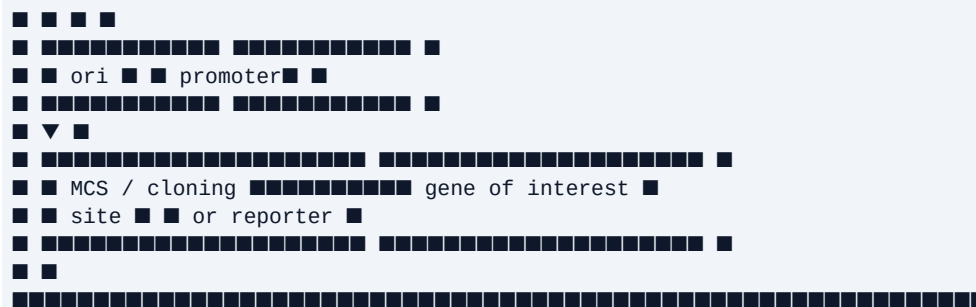


Figure 2.1 — Plasmid architecture (schematic). A real map lists sizes, positions and orientations of every element.

1.1 Expression cassette

An **expression cassette** is the complete transcriptional unit for one expressed sequence: **promoter** → **5' regulatory elements** → **coding sequence (± tags)** → **terminator/polyA signal**. Vectors may carry one cassette or several. When people say "the construct", they usually mean the vector with a complete expression cassette plus insert.

1.2 Fusion proteins

If the insert is cloned **in-frame** with a tag or reporter coding sequence, translation produces a single polypeptide — a **fusion protein** (e.g., GFP-fusion, His-tagged enzyme, lacZ-fusion). Design considerations:

- **Reading frame:** the insert must join the tag without frameshift.
- **Linker:** flexible linkers (e.g., Gly–Ser repeats) preserve independent folding of both parts.
- **Termini matter:** N-terminal fusions need the tag's start codon to dominate; C-terminal fusions must not introduce a premature stop codon before the tag.
- **Function risk:** a fusion can alter the fused protein's localization or activity — always interpret with a control.

2. The complete molecular-cloning workflow

Identify target gene → Design construct → Obtain/amplify DNA → Prepare vector
 → Assemble insert+vector → Introduce into host → Select transformants
 → Screen colonies → Validate construct → Sequence confirmation → Use construct

For each stage: **purpose** → **principle** → **inputs** → **outputs** → **common errors** → **validation**.

Stage 1 — Identify target gene

- **Purpose:** choose the DNA to work with.
- **Principle:** the gene exists in a reference genome/transcriptome database (NCBI, Ensembl).
- **Inputs:** biological question; database record.
- **Outputs:** chosen coding sequence or regulatory region; accession numbers recorded.
- **Common errors:** wrong isoform (alternative splicing); wrong species strain; confusing gene symbol orthologues across species.
- **Validation:** cross-check nucleotide vs protein record; BLAST to confirm identity.

Stage 2 — Design construct

- **Purpose:** decide exactly what the final plasmid looks like before touching a pipette.
- **Principle:** construct design determines everything downstream — enzyme choice, primers, expression behavior.

- **Inputs:** target sequence; vector maps; cloning software (e.g., SnapGene, Benchling, ApE — *examples, not endorsements*).
- **Outputs:** annotated construct map; primer sequences; assembly plan.
- **Common errors:** illegal sites (restriction site inside insert when using restriction cloning); wrong reading frame; missing stop codon for C-terminal tag; promoter incompatible with host.
- **Validation:** in-silico simulation of the whole assembly before ordering anything.

Stage 3 — Obtain/amplify DNA

- **Purpose:** get physical DNA molecules: insert and vector.
- **Principle:** PCR amplifies the insert from template (cDNA, genomic DNA, another plasmid) with primers that add any needed ends; vectors come from plasmid preps or commercial sources.
- **Inputs:** template DNA; primers; polymerase; vector stock.
- **Outputs:** purified PCR product; digested/prepared vector.
- **Common errors:** primer-dimers; wrong annealing temperature; mutations introduced by high-cycle PCR (use high-fidelity polymerase for cloning); contamination.
- **Validation:** agarose gel — a single band of expected size.

Stage 4 — Prepare vector

- **Purpose:** open or linearize the vector so it can accept the insert; or provide compatible/overlapping ends.
- **Principle:** restriction digestion, PCR linearization, or dephosphorylation (see Module 4).
- **Inputs:** vector DNA; enzymes.
- **Outputs:** linearized/processed vector, gel-purified.
- **Common errors:** incomplete digestion; self-ligation background; small excised fragment not removed and then re-ligating.
- **Validation:** gel — clean linear band of correct size; negative control ligation (vector alone).

Stage 5 — Assemble insert + vector

- **Purpose:** join insert and vector into a recombinant molecule in vitro.
- **Principle:** depends on method — ligase-mediated (restriction/TA/blunt), exonuclease-based (Gibson), type IIS (Golden Gate), recombinase (Gateway). Module 4 covers each.
- **Inputs:** insert; vector; assembly enzymes/conditions.
- **Outputs:** mixture containing the desired recombinant product among side products.
- **Common errors:** wrong molar ratios; incompatible ends; degraded enzyme mix.
- **Validation:** none at this stage — a plate-to-plasmid process; validate later.

Stage 6 — Introduce into host

- **Purpose:** put recombinant DNA into cells that will amplify it.
- **Principle:** chemical transformation (competent *E. coli* + heat shock) or electroporation; other hosts need transfection/infection (Module 6).
- **Inputs:** assembly product; competent cells.
- **Outputs:** cells carrying a mixture of recombinant and background plasmids.
- **Common errors:** cells not actually competent (old stock, wrong buffer); DNA dirty (salt kills electroporation).
- **Validation:** positive-control transformation (known plasmid) and negative control (no DNA).

Stage 7 — Select transformants

- **Purpose:** let only vector-carrying cells form colonies.
- **Principle:** plate on the antibiotic matching the vector marker; untransformed cells die/fail to grow.
- **Inputs:** transformed cells; selective plates.
- **Outputs:** colonies — candidates, not confirmations.
- **Common errors:** wrong antibiotic; plates older than the marker tolerates; using too much/too little plasmid for electroporation (arcing).
- **Validation:** compare colony number on positive vs negative control plates.

Stage 8 — Screen colonies

- **Purpose:** distinguish colonies carrying the *desired recombinant* from those carrying empty vector or by-products.
- **Principle:** colony PCR across the insertion site; blue-white screening; restriction mini-prep digest; fluorescence if the insert carries a reporter.
- **Inputs:** colonies.
- **Outputs:** shortlist of candidate clones.
- **Common errors:** screening too few colonies; PCR design that cannot distinguish insert orientation.
- **Validation:** include both a positive-insert-size band expectation and an empty-vector expectation.

Stage 9 — Validate construct

- **Purpose:** confirm structure by restriction analysis (mini-prep DNA cut with diagnostic enzymes gives a predicted band pattern).
- **Principle:** the map predicts fragment sizes; the gel either matches or it does not.
- **Inputs:** mini-prep DNA.
- **Outputs:** verified-by-digest clones.
- **Common errors:** choosing enzymes that give an uninformative pattern (one band).
- **Validation:** pattern matches in-silico digest.

Stage 10 — Sequence confirmation

- **Purpose:** read the actual nucleotide sequence of the junctions/insert — the gold standard.
- **Principle:** Sanger sequencing from vector-priming sites and internal primers.
- **Inputs:** plasmid DNA; sequencing primers.
- **Outputs:** chromatograms and assembled sequence.
- **Common errors:** single-primer coverage (missed mutations elsewhere); not checking the *reverse* strand through the junctions.
- **Validation:** assembled contig matches design exactly, including frame and junctions.

Stage 11 — Use the recombinant construct

- **Purpose:** the actual science — expression, transfection, transformation into a new host, reporting, editing.
- **Principle:** the validated plasmid becomes the reagent.
- **Inputs:** sequence-confirmed plasmid.
- **Outputs:** data, protein, or engineered cells/organisms.
- **Common errors:** assuming expression without testing; forgetting to re-sequence after long-term passaging of strains.
- **Validation:** expression/functional assays appropriate to the purpose (Module 5 onward).

3. Reading a plasmid map

Checklist for any map you are handed:

1. **ori** and copy number class.
2. **Selectable marker(s)** — and whether they differ between bacterial and mammalian parts of the vector (shuttle vectors, Module 3).
3. **MCS or assembly scar architecture** — which sites are unique?
4. **Promoter(s)** and their direction (arrow).
5. **Insert position and orientation** — is orientation marked?
6. **Total size** — expected band on a gel after a single-cutter digest.
7. **Priming sites** — where can sequencing primers bind?

Exam tip: A map without orientation arrows is ambiguous; always state orientation explicitly when describing a construct ("insert in the same orientation as the promoter").

4. Self-check questions

1. Which plasmid component would you modify to (a) increase plasmid yield from a mini-prep, (b) reduce burden on host cells expressing a toxic protein, (c) allow selection in both *E. coli* and yeast?
2. Why is a colony on a selective plate a *candidate* and not a *confirmed* clone?
3. Design-stage error: you plan restriction cloning into an MCS but the insert contains an internal EcoRI site. What are your options?
4. What information does Sanger sequencing add beyond a correct restriction-digest pattern?
5. Sketch (from memory) a plasmid containing: high-copy ori, kanR, T7 promoter, His $\blacksquare$ tag, TEV site, GOI, T7 terminator. Label sizes roughly.

Navigation: [↑ Course Home](#) · [← Previous Topic](#) · [→ Next Topic](#)

Module 3 — Cloning Vectors and Selection/Screening

Level: Beginner → Intermediate

Navigation: [↑ Course Home](#) · [← Previous Topic](#) · [→ Next Topic](#)

■ [Related Labs: Lab 02 · Lab 03 · Lab 04](#) · ■ [Data: Cloning data](#) · ■ [Assessment](#) · ■ [Cheat Sheet](#) · ■ [FAQ](#)

Learning objectives

1. Classify vector systems by host, purpose, capacity, and essential components.
2. Distinguish **selection**, **screening**, and **validation**, and explain why all three stages exist.
3. Choose an appropriate selection/screening strategy for a given cloning scenario.
4. Interpret the results of blue-white screening, colony PCR, and diagnostic digests.

1. Vector systems overview

Vector class	Typical host	Purpose	Capacity / components	Strengths	Limitations
General plasmid cloning vector	<i>E. coli</i>	Propagate a fragment; subclone	ori, antibiotic marker, MCS; ~1–15 kb inserts	Simple, cheap, high copy	Limited capacity; no expression features
Expression vector (bacterial)	<i>E. coli</i> expression strains	Make protein (e.g., T7-based systems in DE3 strains)	Strong inducible promoter, RBS, tag, terminator	High yield, cheap protein	Many eukaryotic proteins misfold; inclusion bodies; codon issues
Mammalian expression vector	Transfected mammalian cells	Express genes in eukaryotic context	CMV/EF1α/PGK-type promoters, Kozak, intron, polyA signal, mammalian selectable marker (neo/puro/hygro)	Proper folding, PTMs; transient or stable	Transfection efficiency varies; silencing of integrated DNA
Shuttle vector	Two hosts (e.g., <i>E. coli</i> + yeast, or <i>E. coli</i> + mammalian)	Build in <i>E. coli</i> , use in second host	Two origins, two selectable markers	Flexible pipeline	Larger; more components to verify
Viral vector (conceptual)	Depends on virus (retro/lenti, AAV, adenoviral)	Efficient gene delivery into cells/animals	Envelope/packaging elements, transgene cassette; replication functions usually provided in trans by packaging cells	Excellent delivery, incl. non-dividing cells (AAV, lenti)	Cargo size limits (AAV ~4.7 kb); biosafety level depends on system; production complexity
BAC (bacterial artificial chromosome)	<i>E. coli</i>	Clone very large genomic loci (~100–300 kb)	F-factor-based single-copy ori, selection marker	Stable maintenance of huge DNA; low recombination	Low copy → low DNA yield; harder manipulation
YAC (yeast artificial chromosome)	<i>S. cerevisiae</i>	Clone megabase-scale fragments	Yeast centromere, telomeres, ARS, selection markers	Largest capacity (up to ~1 Mb)	Chimerism/instability; technically demanding

Vector class	Typical host	Purpose	Capacity / components	Strengths	Limitations
Plant transformation vector (binary)	<i>Agrobacterium tumefaciens</i> (+ <i>E. coli</i> build host)	Deliver T-DNA into plant genomes	Binary: T-DNA border repeats flanking the cassette on one plasmid; vir genes on a helper plasmid (or integrated in disarmed <i>Agrobacterium</i> strain)	Native plant transformation machinery	Plant tissue culture/regeneration is species-specific and slow

Notes:

- "Examples of common architectures (pUC-, pBR322-, pET-, CMV-driven mammalian, binary Ti-derived)" are *archetypes* — thousands of derivatives exist. Consult the actual map of the vector you use.
- BACs and YACs matter for genomics: sequencing projects, transgenic loci with full regulatory context, and synthetic-genome assembly.

1.1 How capacity shapes use

Capacity (log scale, kb)

plasmids BACs YACs

■■■■■■■■■■ ■■■■■■■■■■ ■■■■■■■■■■ ■■■■■■■■■■

1–15 kb ■ 100–300 kb ■ up to ~1000 kb

genes, ■ gene loci ■ genome regions,

cassettes ■ with ■ synthetic genomes

■ regulatory ■

■ context ■

Figure 3.1 — Vector capacity ranges (approximate, log scale).

2. Selection vs screening vs validation

These words are not synonyms. Confusing them causes experimental design errors.

Concept	Definition	Acts on	Examples
Selection	A condition under which undesired cells cannot grow — population-level enrichment	Whole colonies/cells	Antibiotic resistance marker; auxotrophic complementation (host missing a gene survives only if plasmid supplies it)
Screening	A test applied per candidate to distinguish wanted from unwanted clones	Individual colonies	Blue-white; colony PCR; mini-prep digest; fluorescence
Validation	Definitive confirmation of identity/sequence/function	Candidate clones	Sanger sequencing; functional assay; expression check

Transformation

↓

Selection ← only vector-carrying cells grow (marker)

↓

Candidate colonies ← hundreds/thousands

↓

Screening ← colony PCR / blue-white / digest → shortlist

↓

Molecular validation ← diagnostic digest pattern matches map

↓

Sequence confirmation ← Sanger across junctions + full insert

↓

Functional validation ← expression/assay appropriate to purpose

Figure 3.2 — Why every stage is needed: selection removes cells without vector; screening removes clones with the wrong vector; validation proves the surviving clone is what you designed.

2.1 Selectable markers in detail

- **Antibiotic resistance:** the workhorse. Ampicillin/carbenicillin, kanamycin, chloramphenicol, tetracycline, spectinomycin — each pairs with a resistance gene; concentrations are *institutional/protocol-specific* and not universal.
- **Auxotrophic selection:** the host lacks a biosynthetic gene (e.g., *hisB* or *ura3* deficiency); growth on minimal medium requires the plasmid-borne copy. Antibiotic-free alternative used in some industrial and yeast contexts.
- **Mammalian markers:** neo/G418, puromycin, hygromycin, blasticidin — used to derive stable cell lines.
- **Counter-selection tools** (conceptual): markers that *kill* cells carrying them under specific conditions (e.g., *sacB* on sucrose in many Gram-negatives; HSV-TK + ganciclovir in mammalian cells) — essential for scarless engineering workflows.

2.2 Screenable markers and blue-white screening

Blue-white screening exploits **lacZ α -complementation**: the host strain carries a deletion of lacZ α -domain; the vector supplies lacZ α . On X-gal, an empty vector yields **blue** colonies (functional β -galactosidase); an insert disrupting lacZ α yields **white** colonies. White = candidate insert, but insert presence $\neq$ correct insert — screen further.

Other screenable readouts:

- **Fluorescent reporters** (GFP etc.): direct visualization; also enable FACS sorting.
- **Colorimetric reporters:** lacZ/X-gal anywhere in a cassette; alkaline phosphatase histochemistry.
- **Positive-selection cloning:** vectors engineered so that insertion *restores* a lethal function or *disrupts* a toxic gene (ccdB-type systems) — reduces empty-vector background.

2.3 Molecular screening tools

Colony PCR — pick a colony into PCR mix with primers flanking the insertion site. Interpretation:

```
empty vector correct insert wrong-orientation insert
(small band) (insert-size band) (bands differ)
■ ■ ■
▼ ▼ ▼
~200 bp ~1.5 kb diagnostic pair
```

- Use one vector primer + one insert primer to *force* orientation-specific readout.
- Expect both an empty-vector-size band and an insert-size band possibility — design primers so the two are distinguishable.

Diagnostic restriction digest of mini-prep DNA: choose enzymes that cut asymmetrically so orientation gives different fragment patterns. A single uninformative band is a wasted experiment.

Sanger sequencing — final word for sequence identity, including PCR-introduced mutations.

2.4 Expression validation

Sequence-correct $\neq$ expressing. Confirm at the level your question requires:

- **RNA:** RT-PCR/RT-qPCR against the transcript.
- **Protein:** Western blot (anti-tag or anti-protein), or functional fluorescence.
- **Function:** enzyme activity, rescue phenotype, reporter output — depends entirely on the construct's purpose.

3. Worked example — choosing a strategy

Scenario: You ligate a 1.2 kb insert into a 3.5 kb vector carrying amp^R and lacZ α , and transform.

1. **Selection:** plate on ampicillin — every colony has *some* vector.
 2. **Screen:** white colonies on X-gal are candidates; pick 6–12, colony PCR with a vector primer + insert primer.
 3. **Validate:** mini-prep two PCR-positive clones; digest with EcoRI + HindIII (which flank the MCS); expect 1.2 kb + 3.5 kb bands, not the empty-vector pattern.
 4. **Sequence:** Sanger from flanking vector primers covers the entire 1.2 kb insert and both junctions.
 5. **Function:** if the construct is a promoter-reporter, transiently express in the relevant cell type and compare against promoterless vector and empty-vector controls.
-

4. Self-check questions

1. A transformation on antibiotic plates yields 500 colonies, all of which turn out by colony PCR to be empty vector. Which stage failed, and what are two fixes?
 2. Why does blue-white screening not guarantee an insert of the *correct* sequence or size?
 3. When would you choose a BAC over a standard plasmid? Give a concrete research question.
 4. Distinguish auxotrophic selection from antibiotic selection, and give one situation where auxotrophy is preferable.
 5. Your insert is toxic when expressed even weakly in *E. coli*. Which vector features help (ori copy number, promoter type, inducibility)? Explain.
-

Navigation: [↑ Course Home](#) · [← Previous Topic](#) · [→ Next Topic](#)

Insert:vector molar ratio: ligation kinetics depend on effective concentrations. Typical starting points are 3:1 to 5:1 insert:vector molar ratios for sticky-end ligations (formulas and worked calculations appear in Lab 03 and the Cheat Sheet). Too little insert → empty vector; too much → multiple inserts or concatemers.

Directional cloning: cut the vector with two different enzymes and the insert with the same two → non-identical sticky ends force the insert to ligate in one orientation only.

1.3 Strengths / limitations

- **Strengths:** cheap, well-understood; scarless at the junctions if sites preserved; directional possible.
 - **Limitations:** need unique sites in both molecules; internal-site conflicts; multi-fragment assemblies become combinatorially painful; ligase is sequence-agnostic (background self-ligation).
-

2. TA cloning

Principle: *Taq* polymerase adds a single 3'-A overhang to PCR products (non-template-directed). A linear "T-vector" carries complementary 3'-T overhangs. A-overhangs anneal to T-overhangs; ligation completes the circle.

- **Advantages:** simple, fast, directional at the single-base level; ideal for capturing PCR products directly.
 - **Limitations:** high-fidelity polymerases (Pfu, Q5-class) generate **blunt** products — they require an A-tailing step before TA cloning; orientation not controlled; background from non-specific bands.
-

3. Blunt-end cloning

Principle: ligate blunt-ended insert into blunt-ended vector. Any blunt molecule can ligate to any other, so:

- **Advantages:** universal — works when no convenient sites exist; any PCR product or sheared fragment can be used.
 - **Limitations:** ligation is much less efficient than sticky-end ligation; no directionality; vector self-ligation is the dominant background → dephosphorylation is essentially mandatory; frequent need for high insert excess.
-

4. Gibson Assembly

Principle: an isothermal, one-step, multi-fragment assembly using three activities acting on fragments with ~20–40 bp terminal overlaps:

```
5' exonuclease chew back 5' ends → exposes 3' single-stranded overhangs
↓
annealing overlapping ssDNA ends find each other by sequence
↓
polymerase fills gaps at the annealed junctions
↓
DNA ligase seals remaining nicks
```

- **Strengths:** seamless (no restriction-site scars); assembles many fragments in one tube; order and orientation determined by the overlaps themselves; tolerant of awkward sequences (no need for sites).
- **Limitations:** requires designing primers with correct overlaps; repeated sequences cause misassembly; very short or very long fragments each have sweet-spot ranges; cost of enzyme mix.

Multi-fragment design intuition: arrange fragments as a circle; each fragment's end must overlap its neighbor's start. The overlap is the addressing scheme — like matching jigsaw edges.

5. Golden Gate Assembly

Principle: exploit **Type IIS restriction enzymes** (BsaI, BsmBI, Esp3I, SapI...) that cut *outside* their recognition site, leaving **programmable 4-bp overhangs**.

recognition site cut position
GGTCTC(N1) ▼ ■■■■ NNNN|NNNN ← the 4 overhang bases are
▲ whatever YOU design

- Because the overhang bases are designer-chosen, each fragment–fragment junction can be assigned a unique 4-bp "address".
- **One-pot assembly:** digest and ligate happen simultaneously; as the correct overhang pairs ligate, the recognition sites are consumed (they sit outside the final product), so the reaction becomes directional and self-selecting for the intended product.
- **Strengths:** highly efficient; unlimited (practically, tens of) parts; the core of **modular cloning** frameworks (e.g., MoClo-style hierarchies in plant and synthetic biology) — parts are stored in level-0 plasmids with standardized fusion sites, then combined into transcription units, then multi-gene constructs.
- **Limitations:** every part must be domesticated (internal Type IIS sites removed or silently mutated); scar at each junction is 4 bp (usually benign); designing the fusion-site map requires care; enzymes are temperature-sensitive (assembly often performed with alternating digestion/ligation cycling).

6. Gateway cloning (conceptual)

Principle: site-specific recombination borrowed from bacteriophage λ . Recombination between **att** sites moves a DNA segment ("cassette") between plasmids without cutting/ligating.

attB1-Gene-attB2 + donor vector (attP1...attP2, ccdB■)
■ BP clonase (λ integrase-type activity)
▼
entry clone (attL1-Gene-attL2) ← your gene, archived
■ LR clonase (with destination vector attR1...attR2)
▼
expression clones in many destination vectors

- **Entry clone:** a verified, sequence-confirmed archive of the gene.
- **Destination vector:** any backbone accepting the cassette (different promoters, tags, hosts).
- **Strengths:** one verified entry clone feeds unlimited destination vectors; excellent for parallel expression testing.
- **Limitations:** att-site recombination scars remain; destination-vector background requires counter-selection (commonly ccdB-based); clonase mixes are proprietary and costly; the cassette cannot be freely re-designed the way Gibson/Golden Gate allow.

(This module describes the recombination logic; consult current vendor documentation for any actual system before use — this course does not endorse specific commercial products.)

7. Comparison table

Technique	Principle	Strength	Limitation	Typical application
Restriction–ligation	Endonuclease cuts → cohesive/blunt ends → ligase seals	Cheap, robust, well-trodden	Needs suitable unique sites; background self-ligation; poor for multi-fragment	Single-insert subcloning; directional insert swap
TA cloning	Taq A-overhang ↔ T-vector	Fast capture of PCR products	No orientation control; incompatible with blunt-products without A-tailing	Quick PCR-product archiving

Technique	Principle	Strength	Limitation	Typical application
Blunt-end cloning	Ligase joins flush ends	Universal — any blunt fragment	Low efficiency; no directionality; heavy background	Cloning when no sites available; adaptor-based workflows
Gibson Assembly	5' exonuclease + annealing + polymerase + ligase on overlapping fragments	Seamless; multi-fragment; no site constraints	Overlap design burden; repeats cause misassembly; enzyme mix cost	Synthetic constructs; joining several pieces; vector + 2 inserts
Golden Gate	Type IIS → designer 4-bp overhangs; digest+ligate one-pot	Extremely efficient; modular; multiplex	Domestication of internal sites; 4-bp scars; fusion-site planning	Modular cloning; combinatorial libraries; pathway assembly
Gateway	att-site recombinational transfer	Reuse one entry clone across many vectors	Recombination scars; proprietary; counter-selection background	Parallel expression-vector screening; ORFeome-style projects

8. Choosing a technique — decision guide

Need directional, seamless, multi-fragment?

- yes → Gibson ($\leq$ ~6–8 fragments, custom overlaps)
- or Golden Gate (many parts, modular reuse, standardized parts)
- no →

Do unique restriction sites exist in both molecules?

- yes → restriction-ligation (directional, dephosphorylate vector)
- no →

PCR product capture only? → TA cloning

otherwise → blunt-end cloning (accept lower efficiency)

Parallel reuse across many backbones? → Gateway

9. Self-check questions

1. You must clone a 4.5 kb insert with an internal BamHI site into a vector cut with BamHI. Options?
2. Why does Golden Gate assembly become *more* product-specific as the reaction proceeds, while ordinary ligation does not?
3. Your Gibson assembly of 5 fragments produced colonies whose plasmids lack fragment 3. Name three design-level causes.
4. When is Gateway clearly better than Gibson? When is it clearly worse?
5. Explain why dephosphorylation helps blunt-end cloning far more than it helps a two-enzyme directional sticky-end cloning.

Navigation: [↑ Course Home](#) · [← Previous Topic](#) · [→ Next Topic](#)

Module 5 — Expression Constructs and Reporter Genes

Level: Intermediate → Advanced

Navigation: ↑ [Course Home](#) · ← [Previous Topic](#) · → [Next Topic](#)

■ [Related Labs: Lab 06 — Reporter Gene Analysis](#) · [Lab 08 — Computational Gene Expression](#) · ■ [Data: Gene-expression data](#) · ■ [Assessment](#) · ■ [Cheat Sheet](#) · ■ [FAQ](#)

Learning objectives

1. Identify every regulatory element in an expression construct and predict how changing it changes expression.
2. Choose promoters and reporter systems appropriate to host, question, and output.
3. Design promoter-reporter and fusion constructs and anticipate their interpretation pitfalls.

1. Anatomy of an expression construct

Element	Function	Design notes
Promoter	RNA polymerase binding + transcription initiation	Strength, host specificity, inducibility, tissue specificity are independent axes
Enhancer	Distal, orientation-independent positive regulatory element	Can act from kb away; tissue-specific enhancers are the key to spatial expression studies
Silencer	Distal negative element	Reduces/limits expression domains
Operator (prokaryotic)	Repressor/activator binding site modulating transcription (e.g., lac operator)	Basis of inducible bacterial systems
Ribosome-binding site (Shine–Dalgarno)	Prokaryotic translation initiation — pairs with 16S rRNA	Spacing from start codon matters (classic optimum ~5–10 nt)
Kozak sequence	Eukaryotic translation-initiation context around the AUG (consensus gccRccAUGG)	Strength of the context modulates initiation efficiency
5'/3' UTRs	mRNA stability, localization, translation efficiency	miRNA binding sites; stem-loops; Zipcode elements
Terminator (prokaryotic)	Ends transcription (rho-dependent or -independent)	Prevents read-through into plasmid elements
Polyadenylation signal (eukaryotic)	3' end processing + nuclear export + stability	e.g., BGH or SV40 polyA in mammalian vectors; NOS terminators in plant constructs
Intron (some mammalian vectors)	Often boosts expression; supports proper processing	Placement matters

1.1 Expression modes

Mode	Behavior	When to use
Constitutive	Always on, roughly constant	Housekeeping studies; simple protein production
Inducible	Off (or low) until a signal is added/removed	Toxic genes; timed expression; dosage control (lac/IPTG-, Tet/doxycycline-, Gal4/ER-based systems)
Tissue-specific	On only in particular cell/tissue type	Spatial studies; cell-type-specific reporters; safer in vivo expression
Developmentally regulated	On only at certain stages/states	Developmental biology; stage-specific perturbation

Key concept: swapping one regulatory element can completely change the experiment. Same coding sequence + constitutive promoter = "where does the protein go?" + tissue-specific promoter = "where does this gene normally act?"

2. Reporter genes

A **reporter gene** produces a measurable product under the control of the regulatory element being studied. The reporter converts an invisible regulatory state into a signal: fluorescence, color, or light.

2.1 GFP (green fluorescent protein)

- **Source:** *Aequorea victoria* jellyfish. **Chemistry:** chromophore formed autocatalytically from Ser65–Tyr66–Gly67; requires only O₂ (no exogenous substrate). **Excitation/emission:** wild type has major 395 nm/minor 475 nm excitation, ~508 nm emission; engineered variants (EGFP etc.) improved brightness, folding at 37 °C, monomerization.
- **Measures:** location and/or amount of expression, in live or fixed tissue, non-destructively.
- **Strengths:** live imaging; cell/molecule tracking; fusion proteins; FACS sorting; no substrate needed.
- **Limitations:** requires oxygen; some phototoxicity/bleaching; fluorescence background in some tissues; maturation time delays signal; fusion can perturb the fused protein; not quantitative at low levels without calibration.

2.2 RFP family and mCherry

- Derived originally from *Discosoma* sp. red fluorescent protein (dsRed); engineered monomers include mCherry, mOrange, mTagBFP family, etc.
- **Strengths:** longer wavelengths penetrate tissue better; spectrally separable from GFP → dual-color experiments.
- **Limitations:** earlier variants oligomerized (disrupting fusions); mCherry maturation is slower than EGFP; brightness varies by variant.

2.3 lacZ (β-galactosidase)

- **Source:** *E. coli* lacZ. **Substrates:** X-gal (histochemical blue precipitate), ONPG/CPRG (solution assays).
- **Measures:** promoter/enhancer activity in fixed tissue with single-cell resolution.
- **Strengths:** decades of literature; vivid whole-mount staining; cheap substrate; enzyme amplification gives sensitivity.
- **Limitations:** requires fixed tissue (end-point); endogenous β-gal activity in some tissues; substrate diffusion limits; slower assay than fluorescence.

2.4 Luciferase

- **Firefly luciferase** (+ luciferin, ATP, O₂, Mg²⁺ → photon emission ~560 nm); **Renilla** and other luciferases use different substrates (coelenterazine). **Gaussia princeps** (Gaussia) luciferase is secreted — useful for non-destructive time courses from culture medium.
- **Measures:** quantitative promoter/enhancer activity (light output ∝ expression over linear range).
- **Strengths:** wide dynamic range, quantitative, cheap assays, live-animal imaging with substrate delivery (firefly luciferin crosses membranes/cell walls reasonably).
- **Limitations:** requires substrate (not purely genetically encoded); luminescence is dim per cell — spatial resolution poorer than fluorescence; batch effects if substrate delivery varies; dual-reporter normalization (Firefly/Renilla) needs care.

2.5 Fluorescence vs luminescence

Property	Fluorescence (GFP/RFP)	Luminescence (luciferase)
Signal source	Excitation light → emission	Enzymatic photon emission
Substrate needed	No (autocatalytic chromophore)	Yes (luciferin/coelenterazine)
Quantitation	Semi-quantitative; background issues	Excellent dynamic range
Spatial resolution	Subcellular (microscopy)	Tissue/organism (low resolution)
Live imaging	Yes	Yes, but substrate must reach cells
Multiplexing	Multiple colors	Dual-luciferase pairs

2.6 Reporter design patterns

Promoter-reporter: [promoter/enhancer]■■■[reporter]
question: "when/where is this promoter active?"

Enhancer-reporter: [minimal promoter]■■■[reporter] + [candidate enhancer]
question: "does this sequence act as an enhancer, and where?"

Fusion reporter: [strong promoter]■■■[protein-of-interest]■■■[GFP]
question: "where does the protein localize / when does it appear?"

Controls for reporter work: promoterless-vector control; empty-vector control; known-active (positive) promoter; mutational disruption of the element; normalization for transfection efficiency (dual-luciferase, co-transfected control plasmid); for enhancers, test both orientations and distances (enhancers are orientation- and distance-flexible — that is the point).

Interpretation pitfalls:

- A reporter reports the **regulatory sequence's behavior**, not the endogenous gene's. Missing chromatin context, distal elements, or 3D contacts can make a reporter lie.
- Fusion reporters can alter localization or function — validate against endogenous patterns where possible (in situ, immunostaining).
- Brightness ≠ biology: maturation kinetics and protein stability shape the signal (stable GFP persists after transcription stops; destabilized/degron-tagged reporters read out dynamics better).

2.7 Landmark biological example

Chalfie et al. 1994 expressed GFP in *C. elegans* (touch-receptor neurons using the mec-7 promoter) and in *E. coli* — the demonstration that GFP works as a genetically encoded marker in living animals without exogenous substrate. Combined with Prasher's earlier cloning of the gfp cDNA and Tsien's later color/brightness engineering, this opened live developmental imaging as a routine method. (Nobel Prize in Chemistry 2008: Shimomura, Chalfie, Tsien.)

3. Worked mini-example

Question: Is the zebrafish *myod* enhancer active in adaxial cells (the myogenic precursors)?

- **Construct:** candidate enhancer → minimal promoter → GFP.
- **Experimental group:** enhancer + minimal promoter + GFP.
- **Controls:** minimal promoter + GFP alone (baseline); a myod cDNA promoter-GFP known to label adaxial cells (positive control); mutated enhancer (motif disruption).
- **Expected readout (from published literature on myod regulatory biology):** GFP in adaxial cells for the functional enhancer; absent/weak for minimal-promoter-only and motif-mutated constructs.
- **Interpretation:** enhancer sufficiency in this context; not proof of *necessity* — deletion of the endogenous element (or CRISPRi) would test that.

4. Self-check questions

1. Why can the same gene appear "on" with a CMV-driven construct but silent with its own promoter in the same cell type?
2. You need to measure promoter activity *quantitatively over time* in cultured cells. Which reporter, and which controls?
3. Your GFP fusion protein aggregates. Give three plausible design causes and fixes.
4. Why is a destabilized GFP (e.g., with a degradation tag) the better choice for studying a *transient* developmental wave of expression?
5. In the enhancer-reporter design above, why test the enhancer in both orientations?

Navigation: [↑ Course Home](#) · [← Previous Topic](#) · [→ Next Topic](#)

Module 6 — GMO Generation

Level: Intermediate → Advanced

Navigation: ↑ [Course Home](#) · ← [Previous Topic](#) · → [Next Topic](#)

■ [Related Labs: Lab 09 — Capstone Project](#) · ■ [Data: Cloning data](#) · ■ [Assessment](#) · ■ [Cheat Sheet](#) · ■ [FAQ](#)

Safety note: this module describes *conceptual workflows* only. All procedures involving GMOs, animals, or regulated biological systems must follow institutional biosafety-committee (IBC/ETH) approval, containment rules, and instructor-approved SOPs. This course deliberately omits operational animal- and plant-manipulation parameters.

Learning objectives

1. Define GMO, transgenic, gene-edited, stable vs transient modification with precision (Module 1 recap in applied form).
2. Describe the conceptual workflow for generating genetically modified bacteria, plants, and animals.
3. Distinguish historically important GMOs from current-generation products and research tools.
4. Connect GMO generation to the cloning modules: every GMO begins with a construct.

1. Foundations — what makes an organism "modified"?

An organism counts as genetically modified when laboratory methods have altered its genetic material in a defined, designed way. Sub-types:

Category	Example	Defining feature
Transgenic	Bt cotton carrying bacterial <i>cry</i> genes	Foreign (cross-species) DNA
Cisgenic	Potato with late-blight resistance gene from a wild potato cross	Same-species DNA
Gene-edited	Herbicide-tolerance allele edited by base editing	Targeted edit, may add no foreign DNA
Knockout	<i>apoE</i> mouse (atherosclerosis model)	Gene inactivated
Reporter organism	GFP zebrafish, Brainbow mouse	Visible marker introduced
Transiently modified	Plasmid-transfected HEK293 cells	Expression without integration

Stable vs transient (research context): transfection of a plasmid into cultured cells gives transient expression (days); selection with a mammalian marker (e.g., puromycin) and/or integration yields a stable line. In embryos, injected editing reagents act transiently but the *edit* can be inherited — a distinction that confuses many students: the reagent is transient; the consequence is stable.

2. Microorganisms

2.1 Conceptual workflow (bacterial genetic modification)

```

Construct DNA (Modules 2-5)
↓
Introduce DNA (chemical transformation / electroporation / conjugation)
↓
Select transformants (antibiotic or auxotrophic marker)
↓
Screen (colony PCR, blue-white, reporter signal)
↓
Validate (diagnostic digest → sequencing → expression check)
↓

```

Characterize phenotype (growth, production titer, resistance, reporter)

2.2 Applications

- **Recombinant pharmaceutical proteins:** human insulin (Humulin, approved 1982 — first rDNA drug), growth hormone, factor VIII, erythropoietin (CHO cells rather than bacteria for glycoproteins).
- **Industrial enzymes:** amylases, proteases, lipases for detergents/food; rennet (chymosin) for cheese.
- **Research microbes:** expression strains, knockouts, biosensors.

2.3 Historical vs current

- **Historic milestone:** 1973 Cohen–Boyer recombinant plasmids; 1982 insulin approval.
- **Current:** engineered *E. coli*/yeast producing artemisinic acid (semi-synthetic artemisinin pathway), bio-based materials (e.g., bio-produced 1,3-propanediol), designed biosensors, minimal-genome organisms.

3. Plants

3.1 Conceptual workflow

```

Identify trait + gene
↓
Build construct (cloning modules) – often binary vector for Agrobacterium
↓
Introduce DNA:
• Agrobacterium-mediated transfer (most dicots, some monocots)
• Biolistics / gene gun (DNA-coated particles)
↓
Tissue culture & regeneration (callus → shoots → roots → plant)
↓
Selection (plant-selectable marker: hpt/nptII/bar; or reporter)
↓
Molecular confirmation (PCR, Southern-type confirmation of integration,
copy number, expression RT-qPCR/protein assays)
↓
Phenotypic & agronomic characterization (greenhouse → field trials under permit)
↓
Regulatory review (varies by country)

```

3.2 Agrobacterium-mediated transformation (conceptual)

- *Agrobacterium tumefaciens* naturally transfers a segment of its **Ti plasmid** — the **T-DNA**, flanked by 25-bp **border repeats** — into plant cells; integration into the plant genome.
- Engineering removes oncogenes from T-DNA ("disarmed" systems) and puts the desired cassette between the borders; **vir** genes (on helper plasmid or chromosome) mediate transfer.
- Binary-vector system: *E. coli*-friendly build plasmid + *Agrobacterium* helper functions.
- **Why plants are special:** any adult plant cell can, in principle, regenerate a whole plant (totipotency) — the engineered cell becomes an engineered organism through tissue culture.

3.3 Biolistic transformation (conceptual)

DNA-coated gold/tungsten particles physically delivered into cells; used for species/cultivars recalcitrant to *Agrobacterium* (e.g., some cereals) and for chloroplast transformation. Tends to give multiple-copy insertions.

3.4 Landmark crops (historically important → present)

Crop	Modification	Mechanism	Status/notes
Bt cotton / corn / brinjal	<i>cry</i> genes from <i>Bacillus thuringiensis</i>	Insecticidal Cry proteins active in specific insect gut pH/conditions	First commercialized 1996 (US); transformed pest control economics; resistance evolution documented in some pest populations → managed as part of IPM (refuge strategies, pyramided traits)
Herbicide-tolerant soybean/corn	CP4 EPSPS (glyphosate tolerance) or <i>pat/bar</i> (glufosinate)	Altered target enzyme insensitive to herbicide	Dominant traits since 1996; weed-resistance issues drove herbicide mixtures and new tolerance traits
PRSV-resistant papaya ('Rainbow')	Viral coat-protein gene	Pathogen-derived resistance	Saved Hawaiian papaya industry after 1990s PRSV epidemic; classic public-sector GMO
Golden Rice	Phytoene synthase (<i>psy</i>) + carotene desaturase (<i>crtI</i>)	β-carotene biosynthesis in endosperm	Proof-of-concept 2000 (Ye et al.); GR2E received food-safety approvals (Philippines 2021, among others); nutrition-focused, public-sector, ongoing deployment debates
Gene-edited crops	e.g., high-oleic soybean (Calyno), non-browning mushroom (US)	Targeted knockouts/edits	Regulatory status varies by country; some (US mushroom) bypassed USDA oversight historically

(Statuses change; the science of each mechanism is stable, the regulatory landscape is not. Cite current sources when writing about commercial status.)

4. Animals

4.1 Conceptual workflow

```

Design construct or editing strategy
↓
Embryo manipulation or stem-cell route
• pronuclear microinjection (classic transgenesis)
• ES-cell chimera (mouse classics)
• cytoplasmic injection of editing reagents (current CRISPR era)
• SCNT (somatic-cell nuclear transfer – Dolly route)
↓
Embryo transfer to surrogate
↓
Founder identification (PCR/sequencing for transgene or edit)
↓
Breeding to establish lines; germline transmission confirmed
↓
Phenotyping under animal-care approval

```

This course intentionally provides no operational parameters (injection timing, hormone regimens, surgical details) — these depend on species, jurisdiction, and animal-care protocols.

4.2 Landmark animals

Organism	Milestone	Year(s)	Significance
Mouse	Pronuclear transgenesis ("supermouse", Gordon & Ruddle; Palmiter/Brinster GH mouse)	1981–82	First routine mammalian transgenesis

Organism	Milestone	Year(s)	Significance
Mouse	Embryonic-stem-cell knockouts (Capecchi, Smithies, Evans)	1980s–89	Targeted gene replacement; Nobel 2007
Sheep	Dolly — SCNT from adult somatic cell	1996 (pub. 1997)	Cloning ≠ transgenesis; nuclear reprogramming in vivo
Zebrafish	Tol2/transposon transgenesis; CRISPR F0 screening	2000s–2010s	Vertebrate developmental genetics at scale
Various	GFP reporters (worm, fly, zebrafish, mouse)	1994→	Live developmental imaging
Goat/rabbit/chicken	Biopharming (e.g., antithrombin in goat milk, ATRYN approved 2006/2009)	2000s	Transgenic animals as protein factories
Pig	α -1,3-galactosyltransferase knockout (xenotransplantation research); GalSafe pig (FDA approval 2020)	2000s–2020	Organ-source research
Dog/cat/fish	Decorative/companion transgenics (GloFish)	2003→	Public engagement; regulatory variations

4.3 The cloning–transgenesis distinction

- **Reproductive cloning (SCNT)** copies an existing genome into an embryo — no new DNA designed.
- **Transgenesis** adds designed DNA, usually via embryos or cells.
- They combine: a transgenic somatic cell line can be the nuclear donor for SCNT (how many transgenic livestock were made before CRISPR-era embryo editing).

5. Cross-kingdom comparison

Axis	Bacteria	Plants	Animals
DNA delivery	Transformation/electroporation	Agrobacterium / biolistics	Microinjection / viral / SCNT / stem cells
From modified cell to organism	Colony = organism	Totipotent regeneration	Embryo development / chimeras
Time to animal/plant	Overnight–days	Months (tissue culture)	Weeks–months + breeding
Copy number / insertion control	Plasmid-defined	Variable insertions (or targeted edits)	Random (classic) vs targeted (editing era)
Selection system	Antibiotic/auxotrophy	Plant markers + regeneration	Drug selection (cells) / genotyping (animals)
Regulatory burden	Containment (BSL)	Field-trial permits	Animal-care + biosafety + permits

6. Self-check questions

1. Why does totipotency make plant transformation conceptually simpler than animal transgenesis?
2. A CRISPR injection creates an edit in some embryonic cells but not others. What is this called, and what does it mean for founder characterization?
3. Why did Bt crops make *refuge* strategies part of their deployment?
4. Distinguish SCNT from transgenesis using Dolly vs a GFP mouse as your examples.
5. Trace any one plant GMO from Module 6 back to the cloning modules: which construct elements were needed?

Navigation: [↑ Course Home](#) · [← Previous Topic](#) · [→ Next Topic](#)

A **double-strand break (DSB)** is the sensor that triggers repair. The cell's repair machinery determines the edit:

DSB

[illegible]

▼ ▼

NHEJ HDR

(non-homologous end (homology-directed repair, joining; active uses a donor template; throughout the cell most active in S/G2 cycle)

110

indels (small ins/del) precise sequence from donor
frameshifts → knockout → knock-in, correction

- **Knockout:** NHEJ indels in an early coding exon → frameshift → premature stop → functional null (note: in-frame indels can escape; multiple sgRNAs or exon deletion improve success).
- **Knock-in:** HDR with a donor (ssODN for small edits; longer templates for tags/full genes). HDR is *inefficient* in many primary cells — a core limitation; small-molecule and cell-cycle tricks are active research/protocol areas.
- **NHEJ-based knock-in alternatives:** homology-independent targeted integration (HITI), NHEJ-mediated tag insertion with sgRNAs flanking both donor and locus.

2.1 Base editing and prime editing

- **Base editors** (Komor et al., 2016; Gaudelli et al., 2017): catalytically impaired Cas9 + deaminase (APOBEC/rADAR families) ± glycosylase inhibitor → single-base conversions (C→T, A→G; expanded lineages for other pairs) in a small editing window, **without a DSB**. Limits: editing window position, bystander bases, off-target deamination.
- **Prime editing** (Anzalone et al., 2019): Cas9(H840A) nickase fused to reverse transcriptase, guided by a **pegRNA** that encodes the intended edit; nicks one strand, reverse-transcribes the edit in. Handles all 12 base-to-base changes plus small insertions/deletions without donor templates. Limits: efficiency is cell-type/locus dependent; large edits remain hard.

3. Traditional transgenesis vs genome editing

Axis	Traditional transgenesis	Genome editing
What is added	Foreign construct ($\pm$ regulatory DNA), often at random loci	Change to an endogenous sequence; often no foreign DNA remains
Where it lands	Insertion site largely random (positional effects, silencing)	Defined locus
Multiplexing	Each gene needs its own construct	Multiple sgRNAs in one experiment
Time (animals)	Founder + breeding, months–years	F0 screening possible in weeks (zebrafish/frog)
Regulatory framing	GMO frameworks	Varies by country — some edits classified like conventional mutants
Limits	Expression unpredictable; large cargo OK (BACs/transposons)	Cargo size limits; delivery; off-targets; mosaicism; not all cells editable
Best for	Reporters, overexpression, rescue, integration with big regulatory context	Knockouts, precise corrections, allele swaps, multiplex screens

Honest summary: **transgenesis** remains the right tool when you need to **ADD** a designed expression unit; **editing** is the right tool when you need to **CHANGE** an endogenous sequence. Many modern projects use both.

4. Off-target effects, mosaicism, validation

4.1 Off-targets

- Guides can direct cutting at partially matched sites (especially with 1–3 mismatches in the distal part of the protospacer).
- Mitigation: careful guide choice with off-target scoring tools; high-fidelity Cas variants; truncated guides; ribonucleoprotein (RNP) transient delivery; and — most important — **empirical verification** in the actual system.
- Detection: targeted amplicon deep sequencing of predicted off-targets; unbiased methods (GUIDE-seq, Digenome-seq, CIRCLE-seq, DISCOVER-seq) exist at research level.

4.2 Mosaicism

Editing an embryo after the first division produces animals whose cells carry *different* edits. Consequences:

- Founder genotyping from tail/blood may miss edits present in germline or other tissues.
- Phenotype interpretation requires awareness of tissue-specific allele distributions.
- Mitigation: earlier delivery (cytoplasmic injection at/ before the one-cell stage where the system allows), RNP delivery to limit persistence, and genotyping *multiple tissues* of founders, plus outcrossing and F1 genotyping.

4.3 Validation & genotyping workflow (conceptual)

```

Edit design (guide choice, donor design)
↓
Delivery (transient – RNP/mRNA/plasmid; context-dependent)
↓
Bulk genotyping (amplicon PCR → Sanger/ICE-type deconvolution, or NGS)
↓
Single-clone/animal isolation
↓
Confirm biallelic status / zygosity (deep sequencing, restriction assays)
↓
Check predicted off-targets (top sites)
↓
Outcross to remove reagent/mosaic issues → establish line
↓
Phenotype + confirm protein/function loss (Western, activity, phenotype)

```

5. Real research examples (examples, not protocols)

- **MYBPC3 correction in human embryos** (Ma et al., Nature 2017) — demonstrated homology-directed correction of a cardiomyopathy mutation; notable for the follow-up literature on repair mechanisms and mosaicism in early embryos.
- **CRISPR screens in cells** (Shalem et al., 2014; Wang et al., 2014) — genome-wide knockout screens identify essential genes and drug-resistance factors.
- **Agricultural edits:** edited *SBE* (starch branching enzyme) tomato lines (Li et al., 2018, and successors), waxy corn, disease-resistant wheat (edited *MLO* orthologues, e.g., Wang et al., 2014).
- **Therapeutics:** ex-vivo edited HSCs for sickle-cell disease (Casgevy, approved 2023 — BCL11A enhancer edit); in-vivo base-editing trial for transthyretin amyloidosis (NEJM 2021).

Each example is cited to show *what the technology achieved*; none is a protocol.

6. Self-check questions

1. Why is the PAM essential to Cas9 specificity rather than incidental?
2. A knockout experiment yields a healthy-looking F0 with an in-frame 3-bp deletion. Interpret.
3. Why do base editors eliminate the HDR-efficiency problem but introduce bystander-edit concerns?

4. Design a validation plan (conceptual) for a zebrafish F0 crisprant vs a mouse F0 founder — where do the plans differ and why?
5. Your supervisor asks whether to build a transgenic overexpression line or an HDR knock-in of a point mutation. Draft the decision considerations.

Navigation: [↑ Course Home](#) · [← Previous Topic](#) · [→ Next Topic](#)

3. Morphogens and gradients

Morphogen concept (Turing 1952; Wolpert's "French flag" model, 1969): a signaling molecule produced by a localized source diffuses, creating a concentration gradient; cells respond according to concentration thresholds, producing different fates at different distances.

Source Sink

Fate A Fate A Fate B Fate B Fate C Fate C
(high) (high) (med) (med) (low) (low)

↑ thresholds define boundaries (French-flag model)

Figure 8.2 — Morphogen gradient and threshold responses.

Real mechanisms include diffusion with degradation, planar transcytosis, and cytonemes — the details differ by system; the *threshold-response* logic is shared.

Landmark examples:

- **Shh (Sonic hedgehog)** — ventral neural tube: notochord/floor-plate-derived Shh induces ventral neuronal fates (V3 interneurons, motor neurons, V2, V1...) at decreasing concentrations/durations (dorsal–ventral patterning).
- **Shh** — limb: ZPA (zone of polarizing activity) gradient patterns anterior–posterior digit identity.
- **BMP gradient** — dorsal neural tube; BMP inhibition by organizer-secreted antagonists (Noggin, Chordin, Follistatin) is central to neural induction.
- **Wnt gradient** — A–P patterning, segmentation; canonical Wnt/ β -catenin specifies dorsal organizer in zebrafish/xenopus.
- **Drosophila Bicoid** — maternal mRNA localized anteriorly; translated into an anterior–posterior protein gradient activating *hunchback* and other zygotic targets (Nüsslein-Volhard & Wieschaus put the segmentation hierarchy together in their 1980 Nature mutagenesis screen; Nobel 1995).
- **Retinoic acid** — rostrocaudal patterning of the hindbrain and limb proximodistal patterning (endogenous metabolite, graded by synthesizing (RALDH) and degrading (CYP26) enzymes).

4. Signaling pathways (the recurring cast)

Pathway	Ligand class	Canonical signal transduction (conceptual)	Developmental roles (examples)
Wnt/β-catenin	Wnts	Receptor (Frizzled+LRP5/6) $\rightarrow$ inhibit destruction complex $\rightarrow$ β -catenin enters nucleus $\rightarrow$ TCF/LEF targets	Axis formation, segment polarity, stem-cell compartments
Hedgehog	Shh/Ihh/Dhh	Ptched inhibition $\rightarrow$ Smoothened $\rightarrow$ Gli activators/repressors	Neural-tube DV patterning; limb AP; somite sclerotome
TGF-β / BMP / Nodal / Activin	TGF- β superfamily	Type I/II receptor serine-threonine kinases $\rightarrow$ SMADs $\rightarrow$ nucleus	Mesoderm induction (Nodal), DV ectoderm patterning (BMP), amnion/chorion
Notch	Delta/Jagged	Cell-contact-dependent; cleaved NICD $\rightarrow$ RBPJ targets	Lateral inhibition (neurogenesis), somite boundaries, vascular tip cells
FGF	FGFs	RTK $\rightarrow$ MAPK/PI3K/PLC γ	Mesoderm migration, limb bud outgrowth (AER), neural induction contexts

Pathway	Ligand class	Canonical signal transduction (conceptual)	Developmental roles (examples)
Retinoic acid	Retinoic acid	Nuclear receptors (RAR/RXR) → direct gene regulation	Hindbrain/caudalization; limb PD patterning; somite
JAK/STAT, Hippo, YAP/TAZ	Cytokines/mechanical	STAT/dimeric TFs; mechanotransduction	Growth control, regeneration, organ size

These pathways' outputs converge on **cis-regulatory elements** (Module 10); a cell's *combinations* of active pathways — its "positional information" — drive fate decisions.

5. Model organisms — who does what best

Model	Strengths for developmental genetics	Classic tools/genes
Drosophila melanogaster	Forward genetics at scale; short generation; segmental body plan	Bicoid/Nanos/Caudal; gap/pair-rule/segment-polarity genes; Hox (ANT-C/BX-C); Ubx/abd-A
C. elegans	Invariant cell lineage (Sulston); apoptosis genetics; RNAi uptake	<i>lin-4/let-7</i> (miRNAs), programmed cell death genes (<i>ced-3/4/9</i>), Notch lateral-inhibition
Danio rerio (zebrafish)	External transparent development; large clutches; forward screens; live imaging	<i>no tail</i> (<i>ntl/Tbx6</i>), <i>pax2.1/no isthmus</i> , <i>shh</i> (somites/floor plate), <i>spadetail</i>
Mus musculus (mouse)	Mammalian genetics; ES-cell knockouts; conditional alleles	Hox clusters, Pax6 (<i>Small eye</i>), Shh, Wnt1, Fgf8, Cre-lox systems
Arabidopsis thaliana	Plant development; genetics; short life cycle	<i>WUSCHEL/CLAVATA</i> (stem-cell niche), <i>KNOX</i> genes, <i>STM</i> (<i>shootmeristemless</i>), <i>LFY</i> , ABC-floral model (MADS-box genes)

(Gene names are italicized for genes, roman for proteins; different conventions apply in different communities — e.g., *Drosophila* lowercase for recessive alleles.)

6. Landmark genes and what they taught

- **Hox genes** — conserved colinear clusters specifying A–P identity (Lewis 1978; McGinnis & Krumlauf). **Colinearity:** 3' genes expressed anterior/early; 5' posterior/late. Disruption → homeotic transformations (antennapedia legs where antennae should be).
- **Pax6** — eye-field specification, conserved from flies (*eyeless*) to humans; ectopic expression makes ectopic eyes in *Drosophila* (Halder, Callaerts & Gehring, 1995).
- **Sonic hedgehog** — morphogen for neural tube/limb (above).
- **Wnt1** — midbrain/hindbrain boundary; canonical Wnt workhorse.
- **myoD** — master-regulator sufficiency for myogenesis (above).
- **Bicoid** — first morphogen with defined molecular identity (maternal mRNA gradient).

7. Gene regulatory networks — first pass

A **gene regulatory network (GRN)** is the graph of transcription factors, signaling components, and cis-regulatory elements that specifies a fate or process.

```

Maternal inputs (Bcd/Nos)
■
▼
Gap genes (Hb, Kr, Gt, Kni)
■ (cross-regulation sharpens boundaries)
▼
Pair-rule genes (eve, ftz, hairy)

```

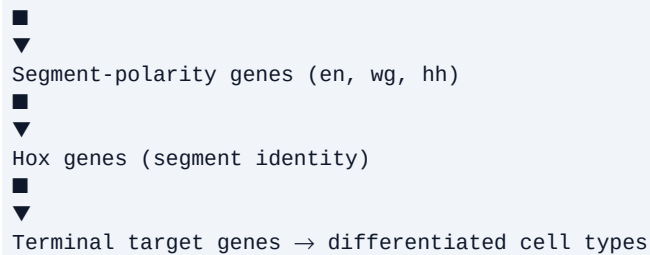


Figure 8.3 — *Drosophila* segmentation GRN hierarchy. Each level feeds the next; cross-regulation within levels sharpens domains.

GRN reasoning rules:

- **Positive feedback** locks a fate in (self-maintaining TF expression).
- **Negative feedback** creates oscillations/refinement (Notch lateral inhibition; somite clock).
- **Cross-repression** creates sharp boundaries (Gt/Kni mutual repression; eve/frtz).
- **Feed-forward loops** filter transient inputs.

8. Self-check questions

1. A cell transplanted from a neural-tube ventral region to a dorsal region *after stage X* still becomes a motor neuron. What property does this illustrate, and when did it arise?
2. Why does a *bicoid* null mother produce embryos without heads — and what does adding anterior bicoid mRNA do (cite the classic result)?
3. Name the pathway and the cis-regulatory logic that makes Notch signaling a "lateral inhibition" device.
4. Why is *myoD* called a master regulator, and what is the caveat to that title?
5. Sketch the DV neural-tube fates as a function of Shh concentration and duration; where do V3, motor neurons, V2, V1 sit?

Navigation: [↑ Course Home](#) · [← Previous Topic](#) · [→ Next Topic](#)

Module 9 — Spatial and Temporal Gene Expression

Level: Intermediate → Advanced

Navigation: ↑ Course Home · ← Previous Topic · → Next Topic

■ Related Labs: Lab 07 — Spatial Gene Expression · Lab 08 — Computational Gene Expression · ■ Data: Spatial data · ■ Assessment · ■ Cheat Sheet · ■ FAQ

Learning objectives

1. Distinguish **where** (spatial), **when** (temporal), and **where+when** (spatiotemporal) expression, and state why each axis matters.
2. Define tissue-specific, cell-type-specific, and developmental-stage-specific expression.
3. Interpret expression gradients and domains in real developmental contexts.
4. Choose methods that resolve space vs time appropriately (bridge to Modules 11–12).

1. Three axes of one question

Every gene-expression statement should locate itself on three axes:

SPATIAL – where in the organism/tissue/cell?
 TEMPORAL – when in development/cell cycle/response?
 QUANTITATIVE – how much, relative to what?

Why the distinction matters: a bulk RNA-seq of a whole embryo integrates over space and time. It can say "gene X rises at stage 12" but cannot say "gene X turns on in the notochord at stage 12, 20 minutes after notochord fate is specified." Different questions require different resolutions.

Concept	Question answered	Example
Spatial expression	Where is the gene expressed?	<i>shh</i> in floor plate/notochord; <i>eve</i> stripes in <i>Drosophila</i> embryo
Temporal expression	When is it expressed?	<i>myoD</i> up-regulation at myogenic commitment
Spatiotemporal	Where and when, jointly?	<i>pax6</i> in eye field from 8-somite stage onward
Tissue-specific	Restricted to a tissue	insulin in β -cells
Cell-type-specific	Restricted to a cell type within a tissue	motor-neuron markers within spinal cord
Stage-specific	Restricted to a developmental window	<i>hunchback</i> early before cellularization
Expression gradient	Continuous concentration change across space	Bicoid A–P gradient; Shh DV gradient
Expression domain	Sharply bounded region of expression	<i>eve</i> stripe 2; <i>engrailed</i> stripes

2. Gradients vs domains — two geometries

GRADIENT (continuous) DOMAIN (bounded)
 conc. conc.
 ▲ ■■■■ ▲
 ■ ■ ■ ■ ■■■■■■■■■■
 ■■ ■■■■ ■ ■ ■
 ■ ■■■■ ■ ■■■■■■ ■■■■■■
 ■■■■■■■■■■■■■■■■ position ■■■■■■■■■■■■■■■■■■■■■ position
 response follows thresholds response is binary, region-limited

Figure 9.1 — Gradients carry positional information continuously; domains are discrete regions (often shaped by cross-regulation on top of gradients).

Biologically, gradients usually *inform*, domains usually *execute*: the Shh gradient informs, the sharply-bounded *olfm/nkx2.2/pax6* domains of the neural tube execute.

3. Spatiotemporal thinking in real examples

3.1 Drosophila anterior–posterior axis

- **Maternal stage:** *bicoid* mRNA localized anteriorly → Bicoid protein gradient (spatial) forms in early nuclear cycles (temporal).
- **Zygotic stage:** Bicoid activates *hunchback*; gap genes form broad domains; pair-rule genes (*eve*, *ftz*) resolve to 7 stripes; segment-polarity genes (*en*, *wg*) to 14 stripes.
- **Takeaway:** each successive layer is both more spatially precise and later in time — patterning is a temporal *refinement cascade* on spatial information.

3.2 Zebrafish somitogenesis

- Somites bud off the presomitic mesoderm (PSM) on a ~30-minute clock.
- **Temporal:** cyclic genes (e.g., *her1/her7*, linked to Notch signaling) oscillate in the PSM.
- **Spatial:** as cells pass the determination front (FGF/Wnt gradient high posterior, low anterior), oscillations freeze into a boundary.
- **Takeaway:** the classic "clock-and-wavefront" — temporal oscillation + spatial gradient = periodic structure. Reading the two axes separately is essential.

3.3 Neural tube DV patterning

- **Spatial:** Shh from notochord/floor plate → ventral gradient; BMPs from roof plate → dorsal gradient.
- **Temporal:** duration and concentration jointly specify class (V3 → motor → V2 → V1 ventrally).
- **Takeaway:** morphogen interpretation depends on *both* concentration and exposure time; a purely spatial view misses this.

3.4 Arabidopsis flower development

- **Spatial:** ABC-model classes occupy concentric whorls (A outer sepals/petals; B petals/stamens; C stamens/carpels).
- **Temporal:** floral meristem identity (LFY/AP1) precedes organ identity; MADS-box expression initiates in defined stages.
- **Takeaway:** radial spatial organization + staged temporal initiation = deterministic organ pattern.

4. Mapping expression — method fit

Method	Spatial resolution	Temporal resolution	Best for
In situ hybridization (chromogenic)	Tissue/cell	Endpoint	Domain mapping, publication figures
In situ (fluorescent, RNAscope-class)	Single-cell, subcellular	Endpoint	Low-abundance transcripts, multiplexing
RNA-FISH	Single-molecule spots	Endpoint	Absolute transcript counts per cell
Immunohistochemistry / IF	Subcellular	Endpoint	Protein-level spatial readout
GFP/lacZ/luciferase reporters	Cell (live)	Live, continuous	Dynamics; enhancer activity
Bulk RT-qPCR / RNA-seq	None (averaged)	Discrete samples	Magnitude, time courses
Single-cell RNA-seq	Cell (lost coordinates)	Snapshot	Cell types, trajectories

Method	Spatial resolution	Temporal resolution	Best for
Spatial transcriptomics (array-based, imaging-based)	Spot (~10–100 μm) or subcellular	Snapshot (time courses by sampling)	Whole-tissue expression maps

(Details and workflows in Modules 11 and 12.)

5. Designing a spatiotemporal study — checklist

1. **State the question in both axes:** "Where is gene X at stage Y?" or "How does the *domain* of X change from stage A to B?"
2. **Choose resolution:** cell type vs tissue region vs whole-embryo.
3. **Choose method(s):** endpoint (in situ, IF) vs live (reporter) vs snapshot-transcriptomic (scRNA/spatial).
4. **Stage control:** developmental staging is the temporal variable — define it rigorously (e.g., zebrafish hpf/somite count; *Drosophila* embryonic stage; E-days for mouse).
5. **Replicates:** biological (clutches/litters), technical (stains/probes).
6. **Controls:** sense probe (or no-probe) for ISH; known marker gene to validate staging/region; reporter-vector controls (Module 5).
7. **Quantification plan:** domain area, intensity, cell counts — decide before imaging.

6. Self-check questions

1. You have bulk RNA-seq showing gene X peaks at 24 hpf in zebrafish. Design the minimal two-experiment follow-up to locate *where*.
2. Why can a gradient and a domain coexist for the same pathway (Shh gradient vs *nkx2.2* domain)?
3. In somitogenesis, which data would convince you a gene is part of the *clock* rather than the *wavefront*?
4. A reporter shows expression in a domain, but in situ for the endogenous gene shows a wider domain. Give two plausible reasons.
5. Which method would you choose to quantify absolute transcript numbers per cell in a tissue? Which for live dynamics?

Navigation: [↑ Course Home](#) · [← Previous Topic](#) · [→ Next Topic](#)

Motif	Wiring	Functional consequence	Example
Feed-forward loop (coherent)	$A \rightarrow B, A \rightarrow C, B \rightarrow C$	Sign persistence; noise filtering	Common in developmental GRNs
Incoherent feed-forward	$A \rightarrow C, A \rightarrow B \dashv C$	Pulse/adaptation	Signaling response dynamics
Lateral inhibition	Delta–Notch between neighbors	Salt-and-pepper fate patterns	Neurogenesis; vulval lineages

3. Enhancer–promoter communication

Enhancers act over large genomic distances; the mechanisms debated and likely multiple:

- **Looping/contact models:** enhancer–promoter physical contact (chromatin loops; TADs organize who can contact whom).
- **Trackability / condensate models:** transcriptional condensates phase-separate components.

Whatever the mechanism, two experimental truths matter:

1. **TAD boundaries constrain enhancer action** — moving an enhancer across a boundary changes its targets (relevant to disease: structural variants can create *neo-enhancers* near oncogenes).
2. **Enhancer activity is context-dependent** — the same enhancer drives different genes in different tissues (Mammalian enhancer–gene maps e.g., from FANTOM5/GeneHancer resources).

4. Chromatin and developmental regulation

- **Histone marks (conceptual):** H3K4me1 with H3K27ac = active enhancer; H3K27me3 = Polycomb-repressed; H3K9me3 = heterochromatin; H3K4me3 = active promoter.
- **DNA methylation:** CpG methylation generally correlates with stable repression; germline/imprinting dynamics; demethylation during early development.
- **Bivalency:** promoters marked with both H3K4me3 and H3K27me3 ("poised") are common in stem/precursor cells — ready for either lineage choice.
- **Pioneer factors:** TFs (e.g., FoxA, Pou5f1/Oct4-class) that bind nucleosomal DNA and open chromatin — why some TFs reprogram cell identity.

5. From data to networks — the inference loop

```

Expression data (scRNA-seq / time course)
+ chromatin data (ATAC-seq, ChIP for TFs/marks)
+ motif information
■
▼
Infer candidate edges (TF → target):
co-expression, motif enrichment in ATAC peaks,
perturbation signatures (KO/KD, CRISPR screens)
■
▼
Prioritize edges → test by perturbation + reporter
(Module 5 constructs; CRISPRi enhancer tests)
■
▼
Validate network function: does perturbing a hub
change the predicted downstream fates?

```

Caveats: co-expression is not causation; inferred edges are hypotheses; the gold standard remains *perturbation* (genetic or pharmacological) with readout at the target level.

6. Worked example — a small developmental GRN

Specification of myoblasts (simplified):

signaling cues (Wnt/Shh/FGF from somite patterning)



Pax3/Pax7 ■■■activates■■■ Myf5 / Myod (commitment)



■ cross-positive ▼

■■■■■■■■■■■■■■■■■■■■ Myogenin (differentiation)



muscle structural genes (Actc1, Myh...)

- Pax3/Pax7 maintain the progenitor state; Myf5/Myod drive commitment; myogenin executes differentiation. Each arrow is supported by knockout/overexpression literature — and each was tested *by perturbation*, not co-expression alone.

7. Self-check questions

1. Why does mutual repression produce sharp boundaries while a single activator gradient does not?
2. What experimental design distinguishes a true feed-forward loop from a coincidental co-expression cluster?
3. Why can an enhancer be validated by CRISPRi more cleanly than by reporter assay? What does the reporter assay still offer?
4. Explain bivalency: why is a "poised" promoter useful in a multipotent cell?
5. In the myogenic network above, which edges would you test first by perturbation, and why?

Navigation: [↑ Course Home](#) · [← Previous Topic](#) · [→ Next Topic](#)

Module 11 — Gene-Expression Experimental Methods

Level: Intermediate → Advanced

Navigation: ↑ [Course Home](#) · ← [Previous Topic](#) · → [Next Topic](#)

■ [Related Labs: Lab 05 · Lab 06 · Lab 07 · Lab 08](#) · ■ [Data: Gene-expression data · Spatial data](#) · ■ [Assessment](#) · ■ [Cheat Sheet](#) · ■ [FAQ](#)

Learning objectives

1. For each method below, state: question answered, input, principle, workflow, output, strengths, limitations, controls, and interpretation.
2. Match method to biological question (resolution, quantitation, live vs endpoint).
3. Distinguish mRNA-level from protein-level readouts and explain why the difference matters.

1. Method matrix (quick reference)

Method	Level	Where/When resolution	Quantitative?	Live?
RT-PCR	mRNA	none	semi	no
RT-qPCR	mRNA	none	yes (relative/absolute)	no
RNA-seq (bulk)	transcriptome	none (averaged)	relative	no
scRNA-seq	transcriptome	cell type (position lost)	relative (sparse)	no
Spatial transcriptomics	transcriptome	tissue spot/cell	relative	no
RNA-FISH / smFISH	mRNA	subcellular	absolute counts	no
In situ hybridization	mRNA	cell/tissue	semi	no
Reporter (GFP/lacZ/luc)	regulatory output	cell/tissue; live	semi-good	yes
Immunohistochemistry/I F	protein	subcellular	semi	no
Western blot	protein	none	semi	no
ELISA	protein	none	yes	no

2. RNA methods

2.1 RT-PCR and RT-qPCR

- **Question:** is transcript X present? How much relative to a reference?
- **Input:** purified RNA (DNA-free); reverse transcription → cDNA.
- **Principle:** reverse transcription + PCR; qPCR monitors amplification in real time (SYBR Green intercalation or TaqMan probe hydrolysis).
- **Workflow:** RNA extraction & QC → DNase → RT → qPCR (technical triplicates) → Ct/Cq values → normalization → $\Delta\Delta C_t$ analysis.
- **Output:** relative expression (fold change) or absolute copies (with standard curve).
- **Controls:**
 - **No-RT control** — detects genomic-DNA contamination.
 - **No-template control** — detects reagent contamination.
 - **Reference genes** (housekeeping) — validated as stable in the actual condition; multiple references recommended (e.g., *GAPDH/ACTB/18S* — but validate!).

- **Inter-run calibrator** for cross-plate comparisons.
- **Common pitfalls:** primer-dimer (melt curves), reference-gene instability across treatments, efficiency $\neq$ 100% ($\Delta\Delta C_t$ assumes $\sim 2\times$ per cycle), RNA degradation (3':5' bias).
- **Interpretation:** fold change is relative to your control; state the reference genes and the control condition explicitly.

2.2 Bulk RNA-seq

- **Question:** which transcripts, at what relative abundance, genome-wide?
 - **Input:** RNA $\rightarrow$ library (fragmentation $\rightarrow$ cDNA $\rightarrow$ adapters $\rightarrow$ indexing).
 - **Principle:** high-throughput short-read (or long-read) sequencing of cDNA; reads map to transcriptome; counts $\propto$ abundance.
 - **Output:** count matrix $\rightarrow$ normalization (TPM for within-sample gene-level expression; DESeq2/edgeR-class models for between-condition testing on raw counts) $\rightarrow$ differential expression, pathway enrichment.
 - **Controls/considerations:** biological replicates (≥ 3); batch structure (randomization, balanced design); strandedness; spike-ins optional; polyA vs total RNA (pre-mRNA inclusion).
 - **Limitations:** averages over cells (heterogeneity masked — the problem scRNA-seq solves); low-abundance transcripts under-sampled; alternative splicing needs deeper depth/long reads.
-

3. Imaging-based methods

3.1 Fluorescence microscopy & reporters

- **Question:** where is a protein (fusion) or where is a regulatory element active (promoter-reporter)?
- **Principle:** genetically encoded fluorophores (GFP family) or immuno-stained fluorophores imaged by epifluorescence/confocal.
- **Controls:** non-transgenic/non-transfected background; spectral bleed-through controls; photobleaching awareness; for fusions — endogenous pattern comparison (in situ/IF).
- **Limitations:** fusion perturbation; autofluorescence; out-of-focus blur in thick tissue (confocal/light-sheet mitigate); maturation kinetics.

3.2 Immunohistochemistry / immunofluorescence (IHC/IF)

- **Question:** where is the *protein*?
 - **Principle:** antibody binding (direct-labeled or secondary-labeled) $\rightarrow$ chromogenic or fluorescent signal.
 - **Controls:** no-primary; isotype control; knockout/known-negative tissue; known-positive tissue; secondary-only.
 - **Limitations:** antibody specificity varies (validate!); fixation/permeabilization artifacts; epitope masking; endogenous pigment/autofluorescence in some tissues.
-

4. RNA localization methods

4.1 In situ hybridization (ISH) — conceptual workflow

```
Sample preparation (fixation, permeabilization)
↓
Probe design (antisense, labeled: DIG/fluor/hapten)
↓
Hybridization (probe base-pairs with target RNA in situ)
↓
Stringent washing (temperature/salt set specificity)
↓
Signal detection (chromogenic alkaline-phosphatase/NBT-BCIP,
```

```

or fluorescence; RNAscope-class signal amplification)
↓
Imaging
↓
Spatial-expression interpretation

```

Probe logic:

- **Antisense probe** — complementary to the transcript → true signal.
- **Sense probe** — same sequence as the transcript → cannot base-pair specifically → *negative control*.
- **Specificity** — determined by probe length/region, hybridization stringency, and organism background; repeated elements must be masked in probe design.
- **Positive control** — a transcript with known, robust spatial pattern (validates the protocol and staging).

Controls summary: sense/no-probe (negative), known-marker probe (positive), staging validation, replicates across clutches/litters.

Strengths/limits: superb spatial mapping at endogenous expression; semi-quantitative at best; endpoint; probe-by-probe optimization; low-abundance transcripts need amplification strategies (RNAscope-class).

4.2 RNA-FISH / smFISH

- Single-molecule sensitivity via many short oligos per transcript → diffraction-limited spots = single transcripts.
- Absolute counts per cell; subcellular localization (e.g., nascent transcription sites, mRNA localization granules).
- Multiplexing (MERFISH/seqFISH-class) scales to hundreds–thousands of genes with barcoded sequential imaging — the bridge between ISH and spatial transcriptomics.

5. Genomic-scale methods

5.1 Single-cell RNA-seq (scRNA-seq) — conceptual

```

Dissociated single-cell suspension
↓
Barcoding/capture (droplet or plate-based; UMI counting)
↓
Library preparation → sequencing
↓
Expression matrix (cells × genes, UMI counts)
↓
QC (filter empty droplets, low-quality/dying cells, doublets)
↓
Normalization → dimensionality reduction (PCA/UMAP)
↓
Clustering → cell-type annotation (marker genes)
↓
Differential expression per cluster;
trajectory/pseudotime (developmental ordering)

```

- **Strengths:** unbiased cell-type discovery; developmental trajectories; rare populations.
- **Limitations:** dissociation loses spatial context; dropout/sparsity; dissociation biases (some cell types lyse); pseudo-time ≠ real time without additional anchors (RNA velocity, lineage tracing).
- **Controls/considerations:** batch design and integration; cell-count balance across conditions; ambient-RNA correction.

5.2 Spatial transcriptomics — conceptual

Two broad families:

- **Array-based** (e.g., Visium-class): tissue on a slide with spotted capture areas (~55 µm spots in the canonical system) — spots capture mRNA with spatial barcodes; after sequencing, each spot carries a spatially-resolved expression profile (spot ≈ several cells).
- **Imaging-based** (MERFISH/seqFISH+/cosMx-class): barcoded probes imaged in situ, iteratively decoded → single-cell/subcellular maps of hundreds of genes.

```

Tissue section on capture surface
↓
Spatially barcoded capture (array) OR barcoded probe imaging
↓
Library prep → sequencing (array) OR image decoding (imaging)
↓
Spatially-resolved expression matrix
↓
Align to histology → tissue-region annotation
↓
Spatial patterns: gradients, domains, ligand-receiver pairs

```

- **Strengths:** keeps spatial context; connects histology with molecular state.
- **Limitations:** lower gene plex (array) or lower throughput (imaging); deconvolution needed when spots contain multiple cell types; resolution–plex trade-offs.

6. Choosing a method — decision guide

Question: WHERE is it expressed? (one/few genes, endogenous)
→ ISH / RNA-FISH / IF (protein)

Question: WHERE + DYNAMICS?
→ reporter (known regulatory element) + ISH validation

Question: WHICH GENES define a cell type/region?
→ scRNA-seq (no coordinates) or spatial transcriptomics

Question: HOW MUCH over time, one/few genes?
→ RT-qPCR time course

Question: GENOME-WIDE, whole tissue, across stages?
→ bulk RNA-seq (+ scRNA/spatial for heterogeneity)

Rule of thumb: sequence/discover broadly → then *validate spatially and functionally* with targeted methods (ISH, reporter, perturbation).

7. Self-check questions

1. Your qPCR shows a 4-fold increase in gene X after treatment, but Western shows no protein change. Give three plausible explanations.
2. Why is the sense probe such a strong negative control — mechanistically?
3. A scRNA-seq cluster expresses two mutually exclusive marker genes. What could this mean (three possibilities)?
4. You need to test whether a candidate enhancer is active in the developing limb. Which method first, and which second for validation?
5. Why can spatial transcriptomics detect ligand–receiver pairs that scRNA-seq on dissociated cells cannot reliably infer?

Navigation: ↑ [Course Home](#) · ← [Previous Topic](#) · → [Next Topic](#)

Module 12 — Data Analysis

Level: Advanced / Computational

Navigation: [↑ Course Home](#) · [← Previous Topic](#) · [→ Next Topic](#)

■ [Related Labs: Lab 02 · Lab 04 · Lab 05 · Lab 08](#) — [Computational Gene Expression](#) · ■ [Data: cloning · gene expression · spatial](#) · ■ [Assessment](#) · ■ [Cheat Sheet](#) · ■ [FAQ](#)

Note: all datasets in this course are **simulated for teaching** — they are clearly labeled, biologically plausible, and not real experimental results. Scripts run with Python 3 + numpy/pandas/matplotlib.

Learning objectives

1. Interpret gel electrophoresis, colony-PCR, and digest data quantitatively.
2. Compute and interpret RT-qPCR relative expression ($\Delta\Delta Ct$) with controls.
3. Perform and interpret a mini bulk-RNA-seq differential-expression workflow.
4. Cluster cells from a mini scRNA-seq matrix and annotate clusters.
5. Read a spatial-expression matrix and identify tissue-specific genes and ligand–receiver structure.

1. Cloning data interpretation

1.1 Gel electrophoresis

Migration $\propto \log(\text{size})$. To size an unknown band, interpolate against the ladder:

```
distance migrated (mm) 10 20 30 40 50
marker sizes (bp) 10000 3000 1000 500 100
log(size) vs distance → linear fit
unknown band at 27 mm → ~1200 bp (from the fit)
```

- **Supercoiled vs linear plasmids:** uncut plasmids migrate anomalously (supercoiled runs faster than its length suggests); always compare digested samples to digested controls or linear markers.
- **Quality flags:** smearing (degradation), trailing (overloading), smiling (voltage too high).

1.2 Colony PCR interpretation

Pattern	Meaning	Action
Insert-size band only	Candidate correct clone	Mini-prep + digest
Empty-vector-size band only	Failed assembly	Check ligation/digest controls
Both bands	Mixed colony or indel	Re-streak single colony
No band	PCR failure or wrong primers	Re-run with positive control DNA

1.3 Diagnostic digest interpretation

Compare the observed fragment pattern to the in-silico prediction:

```
Design: 3.5 kb vector + 1.2 kb insert; digest EcoRI+HindIII
prediction: 3.5 kb + 1.2 kb (correct orientation irrelevant here)
observed: 3.5 + 1.2 → PASS
observed: 4.7 only → empty vector (single site destroyed? check design)
observed: 2.4 + 1.2 + 1.1 → unexpected second site – revisit map
```

(Full worked exercises: [Labs 02 and 04](#); dataset: ``DATA/cloning-data``.)

2. RT-qPCR analysis ($\Delta\Delta Ct$)

Data: Cq values per gene per sample (triplicates), reference gene(s) validated as stable.

1. $\Delta Ct = Cq(\text{target}) - Cq(\text{reference})$
2. $\Delta\Delta Ct = \Delta Ct(\text{condition}) - \Delta Ct(\text{control})$
3. fold change = $2^{(-\Delta\Delta Ct)}$ [assumes ~100% primer efficiency]

Worked example (from `gene-expression-data/qPCR\_timecourse.csv`):

Sample	Cq(target <i>shha</i>)	Cq(<i>ef1α</i>)	ΔCt	ΔΔCt vs 10 hpf	fold change
10 hpf control	24.0	16.0	8.0	0.00	1.0×
12 hpf	24.8	16.0	8.8	0.80	0.57×
18 hpf	26.0	16.0	10.0	2.00	0.25×

(Illustrative numbers; the real CSV has biological replicates and noise.)

Statistical practice: analyze replicates; test ΔCt (or $\Delta\Delta Ct$) with appropriate models rather than comparing fold changes directly; report reference genes, efficiency, and control condition; efficiency-corrected methods (e.g., Pfaffl) when efficiencies differ from 100%.

Common errors: ignoring reference-gene instability; averaging fold changes instead of ΔCt ; treating $n=1$ as $n=3$.

3. Bulk RNA-seq mini-workflow (simulated)

Files: gene-expression-data/rnaseq_counts.csv (genes × samples), metadata.csv (condition, stage).

```
import pandas as pd, numpy as np
from scipy import stats

counts = pd.read_csv("DATA/gene-expression-data/rnaseq_counts.csv", index_col=0)
meta = pd.read_csv("DATA/gene-expression-data/metadata.csv", index_col=0)

# 1. Simple size-factor-style normalization
lib = counts.sum(axis=0)
norm = counts / lib * 1e6 # CPM-style
log = np.log2(norm + 1)

# 2. Differential expression: WT vs KO (3 replicates each)
wt = log[meta[meta.condition=='WT'].index]
ko = log[meta[meta.condition=='KO'].index]
lfc = ko.mean(axis=1) - wt.mean(axis=1)
# (real analyses use DESeq2/edgeR negative-binomial models on raw counts;
# this mini-version teaches the logic)

# 3. Rank and inspect
top = lfc.sort_values()
print(top.head(10)) # down in KO
print(top.tail(10)) # up in KO
```

Interpretation discipline:

- **Fold change vs significance:** sort by $|\log_2 FC|$ but filter by adjusted p-value; a 0.2-log change with $p \approx 0.9$ is noise.
- **Multiple testing:** with 20,000 genes, 5% FDR → ~1,000 "significant" genes by chance — use FDR correction (Benjamini–Hochberg).
- **Effect vs count:** low-count genes have unstable FCs — filter low counts first.
- **Replicates:** $n=3$ is the floor for inference; batch must not confound condition (balanced design).

4. Single-cell analysis (simulated)

Files: gene-expression-data/single_cell_counts.csv (cells × genes), cell_metadata.csv.

```
counts = pd.read_csv("DATA/gene-expression-data/single_cell_counts.csv", index_col=0)
```

```
# QC metrics
lib_size = counts.sum(axis=0)
n_genes = (counts > 0).sum(axis=0)
keep = (lib_size > 500) & (n_genes > 200) # teaching thresholds

# Normalize (CPM) + log
lognorm = np.log2(counts.loc[:, keep] / counts.loc[:, keep].sum(axis=0) * 1e6 + 1)

# Dimensionality reduction: PCA (numpy SVD)
X = lognorm.T.values # cells x genes
Xc = X - X.mean(axis=0)
U, S, Vt = np.linalg.svd(Xc, full_matrices=False)
PCs = U[:, :2] * S[:2] # first two PCs per cell

# Clustering: k-means (sklearn) or simple approach
from sklearn.cluster import KMeans
labels = KMeans(n_clusters=3, n_init=10, random_state=0).fit_predict(PCs)
```

Interpretation:

- Clusters = candidate cell types; annotate by marker genes (e.g., cluster expressing *pax6* high → neural; *myod1* high → muscle).
- **Pseudotime**: order cells along a trajectory (e.g., Monocle-style on PC1) — developmental ordering, *not* clock time.
- **Sparsity**: zeros are mostly dropout — don't interpret absence of a low transcript as absence of biology.

(Full guided exercise: Lab 08.)

5. Spatial-expression analysis (simulated)

Files: spatial-expression-data/spatial_matrix.csv (spots × genes), spot_coordinates.csv (x,y), tissue_annotation.csv (spot → region label).

Exercise logic:

1. Load matrix + coordinates; join annotations.
2. Compute region-mean expression; find genes specific to each region.
3. Plot each marker gene's expression over x,y → in-silico in situ.
4. Test a gradient: correlation of expression with distance from the tissue edge.
5. Ligand–receiver: does the ligand's spatial domain neighbor the receptor's domain? (e.g., ligand in notochord-annotated spots, receptor in adjacent neural-tube spots.)

```
spatial = pd.read_csv("DATA/spatial-expression-data/spatial_matrix.csv", index_col=0)
coords = pd.read_csv("DATA/spatial-expression-data/spot_coordinates.csv", index_col=0)
regions = pd.read_csv("DATA/spatial-expression-data/tissue_annotation.csv", index_col=0)

region_means = spatial.join(regions).groupby("region").mean()
markers = region_means.idxmax(axis=0) # region with max expression
print(markers.value_counts())
```

Interpretation: region-specific genes = spatial markers; expression vs distance plots = gradients; neighboring complementary domains = candidate signaling axes. All *hypotheses* — validate with ISH (Module 11).

6. Building a developmental heatmap and profile

```
import matplotlib.pyplot as plt
# time-course matrix: genes x stages
tc = pd.read_csv("DATA/gene-expression-data/developmental_timecourse.csv", index_col=0)

# z-score genes to compare patterns regardless of magnitude
```

```
z = (tc.sub(tc.mean(axis=1), axis=0)).div(tc.std(axis=1), axis=0)

fig, ax = plt.subplots(figsize=(7, 5))
im = ax.imshow(z.values, aspect="auto", cmap="RdBu_r", vmin=-2, vmax=2)
ax.set_xticks(range(len(tc.columns))); ax.set_xticklabels(tc.columns, rotation=45)
ax.set_yticks(range(len(tc.index))); ax.set_yticklabels(tc.index)
plt.colorbar(im, label="z-scored expression")
ax.set_title("Developmental expression profiles (simulated)")
plt.tight_layout(); plt.savefig("heatmap.png", dpi=150)
```

Reading the heatmap: rows clustered by pattern (early-peak, mid-peak, late-rise); columns = stages. A *maternal* gene starts high and decays; a *zygotic* gene starts at zero and rises. That single distinction is the most important interpretive habit in developmental transcriptomics.

7. Reporting standards for this course

Any quantitative result you report must state:

1. **n** (biological replicates) — technical replicates don't substitute.
 2. **Normalization** (which reference, which size factor).
 3. **Statistics** (test, correction, threshold).
 4. **Effect size** (fold change / log2FC), not just p-values.
 5. **Version of data and code** (in the course: the CSV filenames).
-

8. Self-check questions

1. An uncut plasmid runs "3 kb" on a gel but the map says 5 kb. Explain, and state what to digest with.
 2. Why is it wrong to compute $\Delta\Delta Ct$ per replicate and then average the fold changes?
 3. Your DE analysis finds 1,800 genes at $FDR < 0.05$ with $|\log_2 FC| < 0.1$. What's going on and what do you fix?
 4. In the simulated scRNA data, cluster 3 has *myod1* high but low *actc1*. Propose the cell state and the follow-up experiment.
 5. How would you demonstrate a spatial *gradient* (not just a domain) from the spatial matrix?
-

Navigation: [↑ Course Home](#) · [← Previous Topic](#) · [→ Next Topic](#)

Module 13 — Applications and Case Studies

Level: All levels (integrated)

Navigation: ↑ [Course Home](#) · ← [Previous Topic](#) · → [Next Topic](#)

■ [Related Labs: Lab 09 — Capstone](#) · ■ [Data: all datasets](#) · ■ [Case-Studies assessment](#) · ■ [Cheat Sheet](#) · ■ [FAQ](#)

Twelve detailed case studies, each with: biological question → strategy → mechanism → data/result → interpretation → significance → limitations → discussion questions. **All historical results are cited to their original literature; where a modern status has changed, this is flagged for re-verification.**

Case 1 — Bt crops

- **Question:** Can a bacterial insecticidal protein protect crops with less insecticide spraying?
- **Strategy:** Express *Bacillus thuringiensis* **Cry** genes in crop plants (first commercialized 1996, maize/cotton).
- **Mechanism:** Cry protoxins activate in the alkaline gut of susceptible insects; bind midgut receptors; pore formation → gut damage → death. Specificity arises from receptor + gut environment.
- **Data/result:** Large-scale reductions in insecticide use for target pests documented in agronomic reviews (e.g., meta-analyses of Bt cotton/corn); yield protection under pest pressure.
- **Interpretation:** trait works as designed where pest pressure and susceptibility overlap.
- **Significance:** paradigm of transgenic pest control; reduced broad-spectrum insecticide use.
- **Limitations:** resistance evolution documented in several pests (e.g., *Helicoverpa zea*, *Spodoptera* species) — mitigated by refuge strategies and pyramided toxins; non-target debates (collateral harms to beneficial insects largely associated with insecticide-sprayed non-Bt fields in major meta-analyses, but context matters); gene-flow and seed-saving policy issues.
- **Discuss:** Why do refuges slow resistance? Why is Cry specificity both a strength and a weakness?

Case 2 — Golden Rice

- **Question:** Can rice (staple crop) deliver provitamin A to populations with vitamin-A-deficiency blindness risk?
- **Strategy:** Introduce *psy* (phytoene synthase, daffodil → later maize) + *crtI* (carotene desaturase, *Erwinia*) into rice endosperm.
- **Mechanism:** completes the β -carotene pathway that rice endosperm otherwise stops before.
- **Data/result:** Ye et al. (Science, 2000) proof of concept; GR2E event with improved carotenoid levels; food-safety approvals (Philippines 2021; also US, Australia/NZ, Canada) — **commercial deployment status should be re-verified.**
- **Interpretation:** biofortification technically feasible; efficacy vs actual vitamin-A deficiency depends on intake/deployment.
- **Significance:** flagship public-sector, nutrition-focused GMO; test case for regulatory/public debates.
- **Limitations:** deployment delays; agronomic performance vs local varieties; social acceptance; the alternative interventions (supplementation, diet diversity) comparison is a policy question, not purely scientific.
- **Discuss:** Why was endosperm-specific expression (glutelin promoter) essential? What evidence would convince you it works as public health?

Case 3 — Recombinant insulin

- **Question:** Can human insulin be produced safely and scalably without animal pancreata?

- **Strategy:** Clone human insulin genes/synthetic genes into expression plasmids; produce in *E. coli*/yeast (Humulin approved 1982).
- **Mechanism:** rDNA expression of A/B chains or proinsulin; fermentation; purification; enzymatic conversion (proinsulin route).
- **Data/result:** Humulin; later analogs (lispro, aspart, glargine) engineered for kinetics.
- **Interpretation:** first blockbuster proof that rDNA protein production works clinically.
- **Significance:** established the biopharmaceutical industry; Asilomar-era risk discussions → commercial reality.
- **Limitations:** immunogenicity issues in early products (solved by purification/sequence humanization); production is non-trivial (folding, disulfides).
- **Discuss:** Which expression-host features mattered (no glycosylation need for insulin; proteolytic processing)?

Case 4 — GFP reporter organisms

- **Question:** Can gene expression and protein localization be watched in living organisms?
- **Strategy:** GFP fusions and promoter-GFP transgenics in worm/fly/zebrafish/mouse/plants (Chalfie 1994 →).
- **Mechanism:** genetically encoded fluorophore; promoter or fusion topology determines the readout.
- **Data/result:** entire fields (live developmental imaging, neuron atlases, protein trafficking) enabled.
- **Interpretation:** reporters convert regulatory biology into optics.
- **Significance:** Nobel 2008; the conceptual bridge between this course's cloning module and its expression modules.
- **Limitations:** fusion artifacts; maturation kinetics; phototoxicity; brightness limits in deep tissue.
- **Discuss:** Why is a destabilized GFP better for measuring dynamics than stability?

Case 5 — Transgenic disease models

- **Question:** Can human disease be modeled by engineered genomes?
- **Strategy:** Knockout/knock-in mice (ES-cell era), e.g., *apoE* atherosclerosis; *mdx* (Duchenne); HD knock-in polyQ models; SOD1-G93A ALS.
- **Mechanism:** gene inactivation or disease-allele introduction at endogenous loci.
- **Data/result:** each model reproduces selected aspects of human pathology (never all).
- **Interpretation:** models are tools, not replicas; face/content/predictive validity framing.
- **Significance:** preclinical pharmacology backbone for decades.
- **Limitations:** species differences; single-gene reductionism; poor translation rates for some therapeutic areas.
- **Discuss:** Why do ALS models (SOD1) failed to translate for some therapies that later failed clinically? What does that teach about model limits?

Case 6 — CRISPR knockout

- **Question:** Can genes be knocked out rapidly in any organism?
- **Strategy:** sgRNA + Cas9 delivery to cells/embryos (zebrafish F0 "crisprants", mouse founders, cultured cells).
- **Mechanism:** NHEJ indels at the target → frameshift → null.
- **Data/result:** thousands of papers since 2013; genome-wide screens (Shalem 2014, Wang 2014).
- **Interpretation:** loss-of-function genetics democratized; speed redefined.

- **Significance:** Nobel 2020; routine lab tool.
- **Limitations:** in-frame indels; isoform redundancy; mosaicism in F0; off-targets; compensation.
- **Discuss:** Why does F0 screening work well in zebrafish but less so in mouse (timing of gonadal specification)?

Case 7 — Gene-edited crops

- **Question:** Can crops be improved by editing native genes (no foreign DNA)?
- **Strategy:** knockouts/edits of native loci — e.g., waxy corn (amylose-free via *Wx1* knockout), disease-resistant wheat (*MLO* homoeolog triple knockout), non-browning mushroom (*PPO* edits, USDA not regulated historically), high-oleic soybean oil.
- **Mechanism:** NHEJ/HDR/base editing at native loci.
- **Data/result:** multiple commercial/regulatory milestones across jurisdictions — **regulatory status varies by country and changes; re-verify.**
- **Interpretation:** product-like outcomes without transgene insertion.
- **Significance:** shifts regulatory/public framing in several countries.
- **Limitations:** same trait-improvement constraints as any genetics; off-targets (screened); intellectual property and access questions.
- **Discuss:** Why is *MLO* triple editing in hexaploid wheat a nice illustration of polyploid genetics?

Case 8 — Hox gene regulation

- **Question:** How is body-plan identity encoded along the anterior–posterior axis?
- **Strategy:** decades of genetics + CRMs (e.g., *Hoxd* enhancers), plus ectopic-expression experiments.
- **Mechanism:** colinear clusters; spatial and temporal colinearity; retinoic acid and chromatin dynamics in initiation; long-range enhancers.
- **Data/result:** homeotic transformations from misexpression (Lewis; Antennapedia); enhancer deletions shift domain boundaries.
- **Interpretation:** A–P identity is a transcriptional code read by segmental contexts.
- **Significance:** among the deepest conserved mechanisms in animal development.
- **Limitations:** Hox code is necessary but not sufficient; cofactors (Pbx/Meis) and chromatin context matter.
- **Discuss:** Why does *colinearity* imply an underlying chromatin/regulatory mechanism rather than independent enhancers?

Case 9 — Drosophila developmental patterning

- **Question:** How does a single-cell embryo pattern its A–P axis?
- **Strategy:** saturation mutagenesis screen (Nüsslein-Volhard & Wieschaus, 1980) → hierarchy from maternal → gap → pair-rule → segment-polarity.
- **Mechanism:** Bicoid/Nanos/Caudal maternal gradients activate gap genes; gap-gene cross-regulation; pair-rule stripes; segment-polarity maintenance.
- **Data/result:** mutant phenotypes map to hierarchy levels.
- **Interpretation:** the first "gene regulatory network" understood cell-by-cell.
- **Significance:** Nobel 1995; template for all GRN biology.
- **Discuss:** Why did the screen find *zygot*ic genes but miss *maternal* ones?

Case 10 — Sonic hedgehog spatial signaling

- **Question:** How does one morphogen specify multiple fates in a field?
- **Strategy:** neural-tube DV and limb ZPA experiments; gradient manipulations; cell-culture assays; reporter lines.
- **Mechanism:** Shh gradient → GliA/GliR ratios → concentration- and duration-dependent class specification (V3 → motor → V2 → V1).
- **Data/result:** fate maps align with gradient thresholds; ectopic Shh induces ventral fates dorsally.
- **Interpretation:** French-flag model realized in vertebrates.
- **Significance:** textbook morphogen; clinical relevance (holoprosencephaly; cyclopamine).
- **Limitations:** gradient shape is shaped by feedback, dynamics, and time — not a static morphogen field alone.
- **Discuss:** Why does *duration* of exposure matter in addition to concentration?

Case 11 — Single-cell analysis of development

- **Question:** How many cell types exist in an embryo, and how do they arise?
- **Strategy:** scRNA-seq across developmental stages (e.g., mouse organogenesis atlases; zebrafish embryogenesis; human cell atlas fetal tissue).
- **Mechanism:** per-cell transcriptomes → clusters → marker genes → trajectories (pseudotime, RNA velocity).
- **Data/result:** cell-type catalogs far richer than histology suggested; lineage-informative trajectories.
- **Interpretation:** development as a branching transcriptional landscape.
- **Significance:** redefines the unit of developmental biology from tissue to cell type.
- **Limitations:** lost spatial context (motivates Module 11's spatial methods); dropout; snapshot ≠ dynamics.
- **Discuss:** Which claims *require* spatial validation that scRNA-seq cannot provide?

Case 12 — Spatial transcriptomics in tissue development

- **Question:** How are cell states organized in intact tissue?
- **Strategy:** array-based or imaging-based spatial transcriptomics on developing organs (e.g., brain regions, gut, heart).
- **Mechanism:** spatially barcoded capture or decoded probes → expression maps aligned to histology.
- **Data/result:** region-resolved gene expression; ligand–receiver architecture; gradients preserved.
- **Interpretation:** molecular anatomy at near-cellular resolution.
- **Significance:** Method of the Year 2020 (Nature Methods); bridges histology and omics.
- **Limitations:** resolution vs plexus trade-off; deconvolution needed; cost/throughput.
- **Discuss:** Design the spatial experiment to test the Shh gradient hypothesis end-to-end (Method 11 + data 12).

Cross-case synthesis

Case	Cloning module link	Expression module link
Bt crops	binary vector, promoters, terminators	tissue-specific expression, inheritance
Golden Rice	endosperm-specific promoter	pathway engineering, reporter-like quantification
Insulin	expression vector, host selection	protein production/quality
GFP organisms	promoter-reporter constructs	live spatial-temporal readout

Case	Cloning module link	Expression module link
Disease models	knockouts/knock-ins	expression of disease programs
CRISPR knockout	guide expression constructs	genotype → expression consequence
Edited crops	editing vs transgenesis	trait gene expression
Hox	enhancer-reporter assays	colinear spatial domains
Drosophila	n/a (classical genetics)	GRN cascade spatial refinement
Shh	signaling reporter assays	gradient spatial dynamics
scRNA development	library prep concept	clustering, pseudotime
Spatial	library prep concept	spatial expression matrix

Navigation: [↑ Course Home](#) · [← Previous Topic](#) · [→ Next Topic](#)

Module 14 — Ethics, Biosafety and Regulation

Level: All levels (balanced academic discussion)

Navigation: ↑ [Course Home](#) · ← [Previous Topic](#) · → [Next Topic](#)

■ [Related Lab: Lab 09 — Capstone](#) · ■ [Assessment](#) · ■ [Cheat Sheet](#) · ■ [FAQ](#)

Principle of this module: keep four layers distinct — **scientific evidence** → **risk assessment** → **ethical considerations** → **policy/regulation**. Conclusions at one layer are not evidence at another.

1. Scientific evidence (what we can measure)

- **Molecular containment:** vectors and organisms used for teaching are typically disabled (e.g., non-pathogenic lab *E. coli* strains; disarmed vector systems) — the *system*, not the label, sets risk.
- **Horizontal gene transfer:** possible mechanisms exist (conjugation, transduction, transformation); assessed case-by-case (marker genes, gut environments, selective pressure).
- **Gene flow:** pollen-mediated outcrossing in crops (documented for canola, maize at low frequencies); depends on species, distance, sexual compatibility with wild relatives.
- **Resistance evolution:** documented in target pests for Bt traits (mitigated by refuges/pyramids) and in weeds for herbicide tolerance (selection pressure, not gene flow).
- **Food safety assessment frameworks:** compositional equivalence, allergenicity/toxicity screening of novel proteins (e.g., digestibility, homology searches), feeding studies where indicated — the Codex Alimentarius framework is the international reference.
- **Animal welfare:** both harms (surgical procedures, phenotypes) and potential benefits (disease-resistant livestock) are empirical questions.

Honest summary of evidence: decades of cultivation and consumption have produced a large, mostly reassuring safety literature for the *specific products* assessed through major regulatory systems, alongside *real, documented* challenges (resistance, gene flow, socioeconomic concentration). Blanket "GMOs are safe/unsafe" claims are not scientific statements — product- and trait-specific assessment is.

2. Risk assessment (how we evaluate)

Structured, case-by-case: hazard identification → exposure characterization → consequence estimation → comparison to baseline practice (e.g., current insecticide use). Risk = severity × likelihood × exposure; both *using* and *not using* a technology carry risks (e.g., vitamin-A deficiency vs Golden Rice deployment risk).

Containment levels (conceptual): biosafety levels (BSL-1/2/3/4) classify organisms and procedures by infectivity, transmissibility, and treatment availability; teaching labs with disabled K-12-type *E. coli* and non-viral plasmids typically operate under the lowest containment, but institutional IBC classification governs — *never self-assign*.

3. Ethical considerations (values, not just facts)

- **Autonomy and consent:** labeling and consumer choice; farmers' seed-saving vs IP contracts.
- **Justice and access:** who benefits (smallholders vs corporate licensees); Golden Rice-style public-health framing vs market framing.
- **Animal welfare:** sentience-based limits on modeling and engineering; 3Rs (replacement, reduction, refinement).
- **Germline editing of humans:** heritable edits raise consent-of-future-persons, eugenics-history, and governance-capacity concerns. **The 2018 He Jiankui CCR5 case** is the canonical violation — broadly condemned; moratoria calls and the 2021 WHO governance framework followed.
- **Environmental ethics:** obligations to ecosystems/wild relatives; reversal difficulty (released genes cannot be recalled).

- **Dual-use research of concern (DURC):** gain-of-function studies require oversight frameworks.
- **Genetic privacy:** personal genomic data; discrimination protections (e.g., GINA in the US context).

Course stance: these are legitimate, contested value questions. The university's job is to *equip students to reason* about them — not to install conclusions.

4. Policy and regulation (how societies decide)

Jurisdiction	Framing	Trigger
EU	Process-based	Any technique altering DNA in ways not achievable by conventional means/_mutagenesis exemptions limited_ — new-genomic-techniques (NGT) proposals ongoing to differentiate targeted edits
USA	Product-based (Coordinated Framework: USDA/EP/EPA/FDA by product/use)	Plant-pest risk, pesticide claims, food/feed safety — many edited plants fall outside USDA regulation if no plant-pest DNA used
Argentina/Brazil/Japan/others	Product-like with edit-specific pathways	Approvals for gene-edited crops resembling conventional mutants
International frameworks	Cartagena Protocol on Biosafety (transboundary movements)	Parties' national implementations vary

(Regulatory status of any specific product changes — always verify current status from authoritative sources before writing or citing.)

Public perception: surveys show strong variance by country, product type (medical vs food), and who deploys it (public vs private sector). Trust in *actors* often matters more than trust in *technology*.

5. Risk–benefit reasoning — a worked example

Golden Rice:

- *Scientific:* provitamin A in rice endosperm — demonstrated analytically.
- *Risk:* gene flow to wild rice (low-frequency, studied); food safety assessments completed in several jurisdictions.
- *Ethical:* benefit to vitamin-A-deficient children (real, large) vs autonomy concerns about staple-crop modification; alternative interventions comparison.
- *Policy:* varies by country; Philippines approvals (2021) then legal challenges — a live case.

The layers must not be conflated: safety data does not settle the justice question, and vice versa.

6. Biosafety in the teaching laboratory

Required framing for every lab in this course:

1. **Organisms/materials:** instructor-approved, institutionally-registered strains and constructs only.
2. **Training:** documented lab-safety training before any wet work.
3. **SOPs:** follow your institution's approved SOPs — this course intentionally does not provide operational protocols for anything beyond generic conceptual workflows.
4. **Waste:** dispose of biological waste per institutional biosafety procedures.
5. **Animals/plants/viral vectors/GMO release:** never part of teaching-lab practice without dedicated approval and dedicated facilities.

Simulation vs practice: labs marked *Simulation* or *Computational* involve no living modified organisms; labs marked *Wet Lab (teaching)* use only approved, disabled systems under supervision; labs marked

Demonstration may be instructor-run.

7. Self-check questions

1. Distinguish process-based from product-based regulation with one concrete consequence for a CRISPR-edited crop.
2. Why is "resistance evolution" a *risk-management* problem rather than an argument that the technology fails?
3. Which of the four layers (evidence/risk/ethics/policy) does the labeling debate belong to? Justify.
4. What made the 2018 germline-editing case an ethical failure *independent* of the technical outcome?
5. Give one example where doing nothing carries a larger risk than using the technology — and one where the reverse holds.

Navigation: [↑ Course Home](#) · [← Previous Topic](#) · [→ Next Topic](#)

Module 15 — Advanced Topics

Level: Advanced / Research-oriented (MS and advanced BS)

Navigation: [↑ Course Home](#) · [← Previous Topic](#) · [→ \(end of course — see \[README\]\(../README.md\) for review materials\)](#)

■ [Related Lab: Lab 08 · Lab 09 — Capstone](#) · ■ [Data: single-cell · spatial](#) · ■ [Assessment](#) · ■ [Cheat Sheet](#) · ■ [FAQ](#)

15.1 Enhancer–promoter interactions and 3D genome

- **TADs (topologically associating domains):** self-interacting chromatin regions constraining enhancer search space; boundaries marked by CTCF/cohesin.
- **Loops and enhancer hubs:** CTCF-anchored loops; enhancer clusters contacting multiple promoters; alternation with transcriptional condensate models.
- **Technologies (conceptual):** Hi-C/3C-family for contact maps; Capture-C/4C for locus-specific; ChIA-PET/HiChIP for TF/histone-mark-associated contacts; PLAC-seq/H3K4me3-linked.
- **Disease relevance:** structural variants, enhancer hijacking (e.g., oncogene activation by relocated enhancers), neo-loops.
- **Experimental entry point for students:** CRISPRi of a candidate enhancer + nascent RNA readout (PRO-seq) tests *transcriptional* impact directly.

15.2 Chromatin accessibility and epigenetic regulation

- **ATAC-seq** (and DNase-seq): transposase/endo-nuclease accessibility → open-chromatin map; footprints hint at bound TFs.
- **ChIP-seq / CUT&RUN / CUT&Tag:** histone marks and TF occupancy; H3K27ac for active enhancers, H3K4me1 primed enhancers, H3K4me3 active promoters, H3K27me3 Polycomb-repressed, H3K9me3 heterochromatin.
- **DNA methylation:** bisulfite sequencing; CpG-island promoters; imprinting control regions; global hypomethylation/hypermethylation dynamics in development and cancer.
- **Bivalency in development:** stem/precursor cells keep lineage genes poised (H3K4me3+H3K27me3) — commitment resolves the duality.

15.3 Super-enhancers and transcriptional condensates

- **Super-enhancers:** dense clusters of enhancers marked by exceptionally high H3K27ac/BRD4/Mediator occupancy; often control identity genes (e.g., pluripotency loci, lineage-defining genes).
- **Condensate model:** multivalent interactions among TFs/coactivators/RNAPII form phase-separated transcriptional compartments — a current, still-debated framework (state it as such).
- **Sensitivity:** SE-associated genes are exceptionally sensitive to transcriptional perturbation (e.g., BRD4 inhibition) — therapeutic relevance.

15.4 Transcriptional bursting and noise

- RNAPII transcription is intermittent in many genes — bursts of initiation separated by refractory periods; promoter-proximal pausing and enhancer dynamics shape bursting.
- **Consequences:** cell-to-cell variability in isogenic populations; noise shaping developmental robustness (and its limits).
- **Measurement:** live imaging of MS2/MCP-class stem-loop reporters, single-molecule RNA-FISH distributions (burst size/frequency inference).

15.5 Single-cell heterogeneity and trajectories

- **States vs types:** continuous state variation within a "type"; cell-cycle and stress signatures confound clustering.
- **Pseudotime:** ordering cells along transcriptional similarity; RNA velocity (spliced/unspliced ratios) adds directionality — developmental dynamics from static snapshots.
- **Lineage inference:** mitochondrial/CRISPR barcode lineage tracing; phylogenetic reconstruction of fate decisions.
- **Caveats:** pseudotime $\neq$ chronological time; velocity models assume splicing kinetics stationary within the sample.

15.6 Spatial transcriptomics — frontiers

- Subcellular-resolution platforms; whole-organism-scale atlases (e.g., developing embryo maps).
- **Deconvolution:** spot-level data mixing cell types $\rightarrow$ infer proportions using scRNA references.
- **Cell-cell communication inference:** ligand–receptor co-localization frameworks (e.g., neighboring-domain analysis) — hypothesis-generating.
- **Integration tasks:** aligning scRNA with spatial maps; mapping dissociated atlases back onto anatomy.

15.7 Synthetic gene circuits

- **Motifs to circuits:** toggle switches, repressilators, AND-gate promoters, band-pass filters.
- **Mammalian tools:** inducible systems (Tet-On/Off), recombinase logic (Cre/Flp), synNotch (custom ligand $\rightarrow$ transcriptional program).
- **Design constraints:** burden, orthogonality of parts, insulation, evolutionary stability.
- **Applications:** cell therapies (synNotch-based recognition), biosensors, developmental-pattern engineering (reaction–diffusion-inspired circuits).

15.8 Conditional and tissue-specific expression systems

System	Logic	Use
Cre-loxP	Site-specific recombination at loxP sites; tissue-specific Cre driver $\times$ floxed allele	Conditional knockouts/knock-ins (mouse standard)
Flp-FRT	Analogous to Cre-lox	Complementary labeling/inversion
Dre-rox	Orthogonal recombinase pair	Multiplex logic
Tet-On/Off	Doxycycline-controlled transactivator	Temporal control in cells/animals
Gal4/UAS (fly, zebrafish)	Yeast TF drivers expression from UAS	Tissue-specific driver lines; enhancer trap collections
GAL4-ER² / heat-shock promoters	Ligand/temperature gating	Inducible developmental studies

- **Intersectional strategies:** combining Cre + Flp (or split recombinases) restricts expression to overlapping domains — precision in dense tissues.

15.9 CRISPR beyond knockout

- **CRISPRa:** dCas9-VP64/p65/Rta-class or dCas9-SunTag recruit activators to promoters $\rightarrow$ targeted up-regulation (without genome change) — ideal for testing sufficiency.
- **CRISPRi:** dCas9-KRAB represses transcription (chromatin-based silencing) — ideal for testing necessity, especially of enhancers (CRISPRi of enhancers affects linked promoters with less steric hindrance than promoter targeting).
- **Base editing:** CBE (C $\rightarrow$ T), ABE (A $\rightarrow$ G), plus newer CGBE/GBE variants — DSB-free single-base edits; bystander-edit and off-target deamination remain key caveats.

- **Prime editing:** nickase + reverse transcriptase + pegRNA — search-and-replace edits (all 12 base changes, small indels) without donor templates; efficiency locus- and cell-dependent.
- **Multiplex engineering:** multiple sgRNAs (Cas12a processing arrays; tRNA/ribozyme-flanked guides) — pathways, polygenic traits, combinatorial screens.
- **Epigenome editing:** dCas9-DNMT3A/TET1 — targeted methylation changes; memory of epigenetic edits varies.

15.10 Research-planning checklist (synthesis)

1. QUESTION: mechanistic, specific, falsifiable
2. SYSTEM: model organism / cell type — justified by the question
3. READOUT: matches the question's resolution (space? time? quantitation?)
4. PERTURBATION: knockout vs knockdown vs overexpression vs CRISPRi/a — necessity vs sufficiency
5. CONTROLS: genetic (wild-type, rescue), technical (probe/stain), staging, batch
6. VALIDATION: independent method (ISH validates scRNA; Western validates IF...)
7. QUANTIFICATION: pre-registered plan (what counts as a positive)
8. STATISTICS: n, test, FDR — before data collection
9. ETHICS/BIOSAFETY: approvals, containment, welfare
10. REPRODUCIBILITY: code/data sharing, versioned reagents

Self-check (integrative) questions

1. Design a CRISPRi experiment to test whether a candidate enhancer drives *myod* expression in somites — include the readout and controls.
2. Why does RNA velocity need spliced *and* unspliced counts, and what assumption could break it in a developmental sample?
3. A structural variant moves a limb enhancer 500 kb closer to an oncogene. Predict the consequence and name the technology that would have detected the new loop.
4. Explain how synNotch differs conceptually from a Tet-On system (input → output mapping).
5. Which two technologies from this module would you combine to build a tissue-specific, temporally inducible reporter in zebrafish? Sketch the cross.

Navigation: ↑ [Course Home](#) · ← [Previous Topic](#) · → (course complete — proceed to [\[Assessment\]\(../ASSESSMENT/MCQs.md\)](#) and [\[Capstone Lab\]\(../LAB/Lab-09-Capstone-Project.md\)](#))

References

References and Scientific Sources

Citation policy for this course: only textbooks, landmark papers, reviews and databases that are well-established and verifiable are listed. **No fabricated citations, DOIs, or results.** Students must verify current regulatory/product statuses independently — such statuses change.

Textbooks (authoritative, multi-edition)

- **Alberts, B., et al.** *Molecular Biology of the Cell*. W. W. Norton. (Latest edition; chapters on recombinant DNA, gene expression, development.)
- **Lodish, H., et al.** *Molecular Cell Biology*. W. H. Freeman.
- **Watson, J. D., et al.** *Molecular Biology of the Gene*. Pearson/Cold Spring Harbor.
- **Lewin, B.** *Lewin's Genes* (Lewin's Genes XII+). Jones & Bartlett.
- **Gilbert, S. F., & Barresi, M. J. F.** *Developmental Biology*. Oxford University Press/Sinauer.
- **Wolpert, L., et al.** *Principles of Development*. Oxford University Press.
- **Brown, T. A.** *Gene Cloning and DNA Analysis: An Introduction*. Wiley-Blackwell. (Standard cloning-methods primer.)

Landmark papers (verify via PubMed/DOI when citing)

Recombinant DNA and cloning

- Cohen, S. N., Chang, A. C. Y., Boyer, H. W., & Helling, R. B. (1973). Construction of biologically functional bacterial plasmids in vitro. *PNAS*, 70(11), 3240–3244.
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Genome sequencing and genomics

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CRISPR and genome editing

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- Shalem, O., et al. (2014). Genome-scale CRISPR-Cas9 knockout screening in human cells. *Science*, 343(6166), 84–87.
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- Gaudelli, N. M., et al. (2017). Programmable base editing of A·T to G·C in genomic DNA without DNA cleavage. *Nature*, 551, 464–471. (ABE.)
- Anzalone, A. V., Randolph, P. B., Davis, J. R., et al. (2019). Search-and-replace genome editing without double-strand breaks or donor DNA. *Nature*, 576, 149–157. (Prime editing.)
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- ISAAA Briefs — GM Approval Database (compilation of regulatory statuses; verify current entries).
- Tabi, I., et al. / general spatial-omics method reviews — search "*spatial transcriptomics review*" in *Nature Reviews Methods Primers* (2022) for a current primer.

Databases and authoritative resources

- **NCBI** — GenBank, PubMed, RefSeq, BLAST: <https://www.ncbi.nlm.nih.gov>
- **EMBL-EBI** — Ensembl, Expression Atlas, PRIDE: <https://www.ebi.ac.uk>
- **Ensembl** genomes: <https://www.ensembl.org>
- **UniProt** protein database: <https://www.uniprot.org>
- **ZFIN** (zebrafish): <https://zfin.org> · **MGI** (mouse): <http://www.informatics.jax.org> · **FlyBase**: <https://flybase.org> · **WormBase**: <https://wormbase.org> · **TAIR** (Arabidopsis): <https://www.arabidopsis.org>
- **WHO** — human genome editing governance framework (2021): <https://www.who.int>
- **Codex Alimentarius** — food-safety assessment texts: <https://www.fao.org/fao-who-codexalimentarius>
- **GeneCards / GeneHancer** — human gene–enhancer maps: <https://www.genecards.org>
- **NIH Guide** — biosafety resources: <https://osp.od.nih.gov/biotechnology/>

Course-internal sources

- All course data: DATA/ (simulated; generator DATA/generate_datasets.py).
- Diagrams: DIAGRAMS/ (schematic teaching illustrations; generate_diagrams.py).
- Where this course states regulatory or product statuses, students must re-verify from the sources above before citing.

(Reference count: ~55 entries across 5 categories. If citing in academic work, always confirm volume/page/DOI from the publisher or PubMed — this list deliberately avoids DOIs to prevent transcription errors.)